PART 4

BIG DATA ANALYTICS

Traditional applications of relational databases are based on structured data and they deal with data from a single enterprise. Modern data management applications often need to deal with data that are not necessarily in relational form; further, such applications also need to deal with volumes of data that are far larger than what a single traditional organization would have generated. In Chapter 10, we study techniques for managing such data, often referred to as Big Data. Our coverage of Big Data in this chapter is from the perspective of a programmer who uses Big Data systems. We start with storage systems for Big Data, and then cover querying techniques, including the MapReduce framework, algebraic operations, steaming data, and graph databases.

One major application of Big Data is data analytics, which refers broadly to the processing of data to infer patterns, correlations, or models for prediction. The financial benefits of making correct decisions can be substantial, as can the costs of making wrong decisions. Organizations therefore make substantial investments both to gather or purchase required data and to build systems for data analytics. In Chapter 11, we cover data analytics in general and, in particular, decision-making tasks that benefit greatly by using data about the past to predict the future and using the predictions to make decisions. Topics covered include data warehousing, online analytical processing, and data mining.

CHAPTER 10

Big Data

Traditional applications of relational databases are based on structured data, and they deal with data from a single enterprise. Modern data management applications often need to deal with data that are not necessarily in relational form; further, such applications also need to deal with volumes of data that are far larger than what a single enterprise would generate. We study techniques for managing such data, often referred to as Big Data, in this chapter.

10.1 Motivation

The growth of the World Wide Web in the 1990s and 2000s resulted in the need to store and query data with volumes that far exceeded the enterprise data that relational databases were designed to manage. Although much of the user-visible data on the web in the early days was static, web sites generated a very large amount of data about users who visited their sites, what web pages they accessed, and when. These data were typically stored on log files on the web server, in textual form. People managing web sites soon realized that there was a wealth of information in the web logs that could be used by companies to understand more about their users and to target advertisements and marketing campaigns at users. Such information included details of which pages had been accessed by users, which could also be linked with user profile data, such as age, gender, income level, and so on, that were collected by many web sites. Transactional web sites such as shopping sites had other kinds of data as well, such as what products a user had browsed or purchased. The 2000s saw exceptionally large growth in the volume of user-generated data, in particular social-media data.

The volume of such data soon grew well beyond the scale that could be handled by traditional database systems, and both storage and processing require a very high degree of parallelism. Furthermore, much of the data were in textual form such as log records, or in other semi-structured forms that we saw in Chapter 8. Such data, are characterized by their size, speed at which they are generated, and the variety of formats, are generically called Big Data.

Big Data has been contrasted with traditional relational databases on the following metrics:

- **Volume**: The amount of data to be stored and processed is much larger than traditional databases, including traditional parallel relational databases, were designed to handle. Although there is a long history of parallel database systems, early generation parallel databases were designed to work on tens to a few hundreds of machines. In contrast, some of the new applications require the use of thousands of machines in parallel to store and process the data.
- **Velocity**: The rate of arrival of data are much higher in today's networked world than in earlier days. Data management systems must be able to ingest and store data at very high rates. Further, many applications need data items to be processed as they arrive, in order to detect and respond quickly to certain events (such systems are referred to a *streaming data systems*). Thus, processing velocity is very important for many applications today.
- **Variety**: The relational representation of data, relational query languages, and relational database systems have been very successful over the past several decades, and they form the core of the data representation of most organizations. However, clearly, not all data are relational.

  As we saw in Chapter 8, a variety of data representations are used for different purposes today. While much of today's data can be efficiently represented in relational form, there are many data sources that have other forms of data, such as semi-structured data, textual data, and graph data. The SQL query language is well suited to specifying a variety of queries on relational data, and it has been extended to handle semi-structured data. However, many computations cannot be easily expressed in SQL or efficiently evaluated if represented using SQL.

  A new generation of languages and frameworks has been developed for specifying and efficiently executing complex queries on new forms of data.

We shall use the term Big Data in a generic sense, to refer to any data-processing need that requires a high degree of parallelism to handle, regardless of whether the data are relational or otherwise.

Over the past decade, several systems have been developed for storing and processing Big Data, using very large clusters of machines, with thousands, or in some cases, tens of thousands of machines. The term **node** is often used to refer to a machine in a cluster.

10.1.1 Sources and Uses of Big Data

The rapid growth of the web was the key driver for the enormous growth of data volumes in the late 1990s and early 2000s. The initial sources of data were logs from web server software, which recorded user interactions with the web servers. With each user

clicking on multiple links each day, and hundreds of millions of users, which later grew to billions of users, the large web companies found they were generating multiple terabytes of data each day. Web companies soon realized that there was a lot of important information in the web logs, which could be used for multiple purposes, such as these:

- Deciding what posts, news, and other information to present to which user, to keep them more engaged with the site. Information on what the user had viewed earlier, as well as information on what other users with similar preferences had viewed, are key to making these decisions.
- Deciding what advertisements to show to which users, to maximize the benefit to the advertiser, while also ensuring the advertisements that a user sees are more likely to be of relevance to the user. Again, information on what pages a user had visited, or what advertisements a user had clicked on earlier, are key to making such decisions.
- Deciding how a web site should be structured, to make it easy for most users to find information that they are looking for. Knowing to what pages users typically navigate, and what page they typically view after visiting a particular page, is key to making such decisions.
- Determining user preferences and trends based on page views, which can help a manufacturer or vendor decide what items to produce or stock more of, and what to produce or stock less of. This is part of a more general topic of business intelligence.
- Advertisement display and click-through information. A **click-through** refers to a user clicking on an advertisement to get more information, and is a measure of the success of the advertisement in getting user attention. A **conversion** occurs when the user actually purchases the advertised product or service. Web sites are often paid when a click-through or conversion occurs. This makes click-through and conversion rates for different advertisements a key metric for a site to decide which advertisements to display.

Today, there are many other sources of very high-volume data. Examples include the following:

- Data from mobile phone apps that help in understanding user interaction with the app, in the same way that clicks on a web site help in understanding user interaction with the web site.
- Transaction data from retain enterprises (both online and offline). Early users of very large volumes of data included large retail chains such as Walmart, who used parallel database systems even in the years preceding the web, to manage and analyze their data.

- Data from sensors. High-end equipment today typically has a large number of sensors to monitor the health of the equipment. Collecting such data centrally helps to track status and predict the chances of problems with the equipment, helping fix problems before they result in failure. The increasing use of such sensors to the connection of sensors and other computing devices embedded within other objects such as vehicles, buildings, machinery, and so forth to the internet, often referred to as the **internet of things**. The number of such devices is now more than the number of humans on the internet.
- Metadata from communication networks, including traffic and other monitoring information for data networks, and call information for voice networks. Such data are important for detecting potential problems before they occur, for detecting problems as they occur, and for capacity planning and other related decisions.

The amount of data stored in databases has been growing rapidly for multiple decades, well before the term Big Data came into use. But the extremely rapid growth of the web created an inflection point, with the major web sites having to handle data generated by hundreds of millions to billions of users; this was a scale significantly greater than most of the earlier applications.

Even companies that are not web related have found it necessary to deal with very large amounts of data. Many companies procure and analyze large volumes of data generated by other companies. For example, web search histories annotated with user profile information, have become available to many companies, which can use such information to make a variety of business decisions, such as planning advertising campaigns, planning what products to manufacture and when, and so on.

Companies today find it essential to make use of social media data to make business decisions. Reactions to new product launches by a company, or a change in existing offerings can be found on Twitter and other social media sites. Not only is the volume of data on social media sites such as Twitter very high, but the data arrives at a very high velocity, and needs to be analyzed and responded to very quickly. For example, if a company puts out an advertisement, and there is strong negative reaction on Twitter, the company would want to detect the issue quickly, and perhaps stop using the advertisement before there is too much damage. Thus, Big Data has become a key enabler for a variety of activities of many organizations today.

10.1.2 Querying Big Data

SQL is by far the most widely used language for querying relational databases. However, there is a wider variety of query language options for Big Data applications, driven by the need to handle more variety of data types, and by the need to scale to very large data volumes/velocity.

Building data management systems that can scale to a large volume/velocity of data requires parallel storage and processing of data. Building a relational database that supports SQL along with other database features, such as transactions (which we

study later in Chapter 17), and *at the same time* can support very high performance by running on a very large number of machines, is not an easy task. There are two categories of such applications:

1. *Transaction-processing systems that need very high scalability*: Transaction-processing systems support a large number of short running queries and updates.

   It is much easier for a database designed to support transaction processing to scale to very large numbers of machines if the requirements to support all features of a relational database are relaxed. Conversely, many transaction-processing applications that need to scale to very high volumes/velocity can manage without full database support.

   The primary mode of data access for such applications is to store data with an associated key, and to retrieve data with that key; such a storage system is called a key-value store. In the preceding user profile example, the key for user-profile data would be the user's identifier. There are applications that conceptually require joins but implement the joins either in application code or by a form of view materialization.

   For example, in a social-networking application, when a user connects to the system, the user should be shown new posts from all her friends. If the data about posts and friends is maintained in relational format, this would require a join. Suppose that instead, the system maintains an object for each user in a key-value store, containing their friend information as well as their posts. Instead of a join done in the database, the application code could implement the join by first finding the set of friends of the user, and then querying the data object of each friend to find their posts. Another alternative is as follows: whenever a user u_0 makes a post, for each friend u_i of the user, a message is sent to the data object representing u_i, and the data associated with the friend are updated with a summary of the new post. When that user u_i checks for updates, all data required to provide a summary view of posts by friends are available in one place and can be retrieved quickly.

   There are trade-offs between the two alternatives, such as higher cost at query time for the first alternative, versus higher storage cost and higher cost at the time of writes for the second alternative.[1] But both approaches allow the application to carry out its tasks without support for joins in the key-value storage system.

2. *Query processing systems that need very high scalability, and need to support non-relational data*: Typical examples of such systems are those designed to perform analysis on logs generated by web servers and other applications. Other examples include document and knowledge storage and indexing systems, such as those that support keyword search on the web.

[1] It is worth mentioning that it appears (based on limited publicly available information as of 2018) that Facebook uses the first alternative for its news feed to avoid the high storage overhead of the second alternative.

The data consumed by many such applications are stored in multiple files. A system designed to support such applications first needs to be able to store a large number of large files. Second, it must be able to support parallel querying of data stored in such files. Since the data are not necessarily relational, a system designed for querying such data must support arbitrary program code, not just relational algebra or SQL queries.

Big Data applications often require processing of very large volumes of text, image, and video data. Traditionally such data were stored in file systems and processed using stand-alone applications. For example, keyword search on textual data, and its successor, keyword search on the web, both depend on preprocessing textual data, followed by query processing using data structures such as indices built during the preprocessing step. It should be clear that the SQL constructs we have seen earlier are not suited for carrying out such tasks, since the input data are not in relational form, and the output too may not be in relational form.

In earlier days, processing of such data was done using stand-alone programs; this is very similar to how organizational data were processed prior to the advent of database management systems. However, with the very rapid growth of data sizes, the limitations of stand-alone programs became clear. Parallel processing is critical given the very large scale of Big Data. Writing programs that can process data in parallel while dealing with failures (which are common with large scale parallelism) is not easy.

In this chapter, we study techniques for querying of Big Data that are widely used today. A key to the success of these techniques is the fact that they allow specification of complex data processing tasks, while enabling easy parallelization of the tasks. These techniques free the programmer from having to deal with issues such as how to perform parallelization, how to deal with failures, how to deal with load imbalances between machines, and many other similar low-level issues.

10.2 Big Data Storage Systems

Applications on Big Data have extremely high scalability requirements. Popular applications have hundreds of millions of users, and many applications have seen their load increase many-fold within a single year, or even within a few months. To handle the data management needs of such applications, data must be stored partitioned across thousands of computing and storage nodes.

A number of systems for Big Data storage have been developed and deployed over the past two decades to address the data management requirements of such applications. These include the following:

- **Distributed File Systems**. These allow files to be stored across a number of machines, while allowing access to files using a traditional file-system interface. Distributed file systems are used to store large files, such as log files. They are also used as a storage layer for systems that support storage of records.

- **Sharding across multiple databases.** *Sharding* refers to the process of partitioning of records across multiple systems; in other words, the records are divided up among the systems. A typical use case for sharding is to partition records corresponding to different users across a collection of databases. Each database is a traditional centralized database, which may not have any information about the other databases. It is the job of client software to keep track of how records are partitioned, and to send each query to the appropriate database.
- **Key-Value Storage Systems**. These allow records to be stored and retrieved based on a key, and may additionally provide limited query facilities. However, they are not full-fledged database systems; they are sometimes called NoSQL systems, since such storage systems typically do not support the SQL language.
- **Parallel and Distributed Databases**. These provide a traditional database interface but store data across multiple machines, and they perform query processing in parallel across multiple machines.

Parallel and distributed database storage systems, including distributed file systems and key-value stores, are described in detail in Chapter 21. We provide a user-level overview of these Big Data storage systems in this section.

10.2.1 Distributed File Systems

A **distributed file system** stores files across a large collection of machines while giving a single-file-system view to clients. As with any file system, there is a system of file names and directories, which clients can use to identify and access files. Clients do not need to bother about where the files are stored. Such distributed file systems can store very large amounts of data, and support very large numbers of concurrent clients. Such systems are ideal for storing unstructured data, such as web pages, web server logs, images, and so on, that are stored as large files.

A landmark system in this context was the Google File System (GFS), developed in the early 2000s, which saw widespread use within Google. The open-source Hadoop File System (HDFS) is based on the GFS architecture and is now very widely used.

Distributed file systems are designed for efficient storage of large files, whose sizes range from tens of megabytes to hundreds of gigabytes or more.

The data in a distributed file system is stored across a number of machines. Files are broken up into multiple blocks. The blocks of a single file can be partitioned across multiple machines. Further, each file block is replicated across multiple (typically three) machines, so that a machine failure does not result in the file becoming inaccessible.

File systems, whether centralized or distributed, typically support the following:

- A directory system, which allows a hierarchical organization of files into directories and subdirectories.
- A mapping from a file name to the sequence of identifiers of blocks that store the actual data in each file.

- The ability to store and retrieve data to/from a block with a specified identifier.

In the case of a centralized file system, the block identifiers help locate blocks in a storage device such as a disk. In the case of a distributed file system, in addition to providing a block identifier, the file system must provide the location (machine identifier) where the block is stored; in fact, due to replication, the file system provides a set of machine identifiers along with each block identifier.

Figure 10.1 shows the architecture of the Hadoop File System (HDFS), which is derived from the architecture of the Google File System (GFS). The core of HDFS is

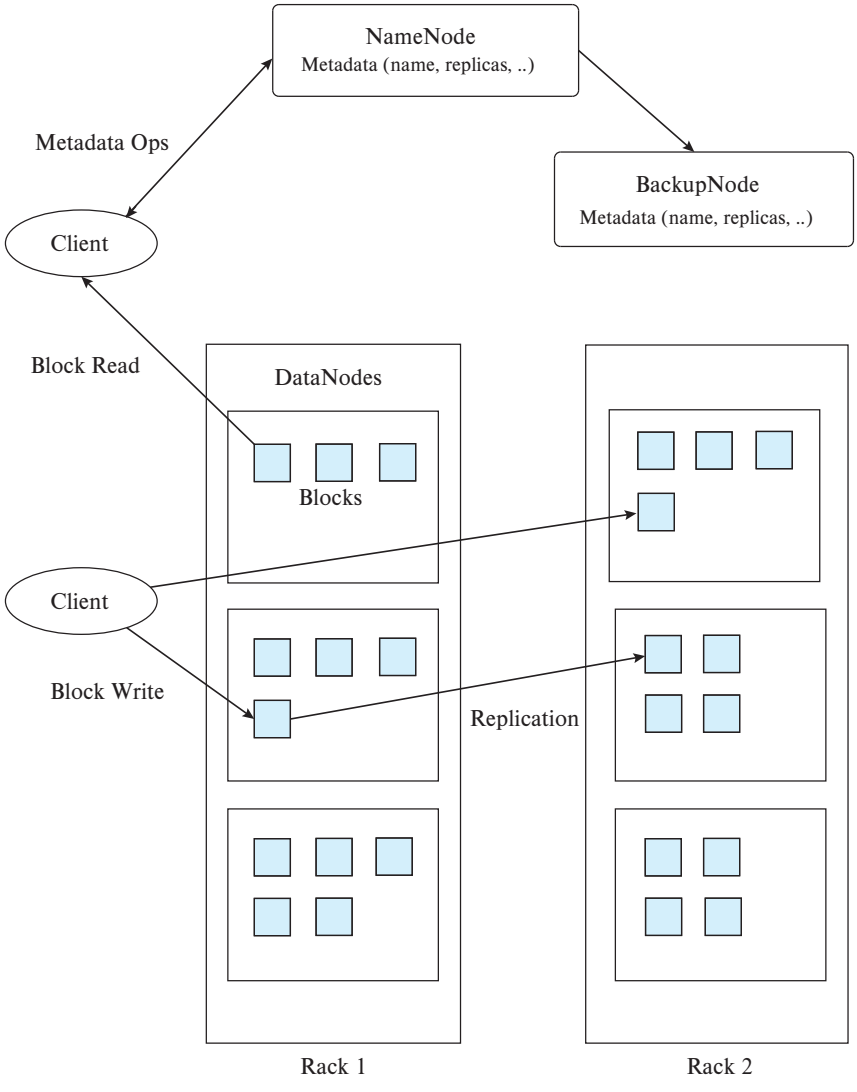


Figure 10.1 Hadoop Distributed File System (HDFS) architecture.

a server running a machine referred to as the **NameNode**. All file system requests are sent to the NameNode. A file system client program that wants to read an existing file sends the file name (which can be a path, such as /home/avi/book/ch10) to the NameNode. The NameNode stores a list of block identifiers of the blocks in each file; for each block identifier, the NameNode also stores the identifiers of machines that store copies of that block. The machines that store data blocks in HDFS are called **DataNodes**.

For a file read request, the HDFS server sends back a list of block identifiers of the blocks in the file and the identifiers of the machines that contain each block. Each block is then fetched from one of the machines that store a copy of the block.

For a file write, the HDFS server creates new block identifiers and assigns each block identifier to several (usually three) machines, and returns the block identifiers and machine assignment to the client. The client then sends the block identifiers and block data to the assigned machines, which store the data.

Files can be accessed by programs by using HDFS file system APIs that are available in multiple languages, such as Java and Python; the APIs allow a program to connect to the HDFS server and access data.

An HDFS distributed file system can also be connected to the local file system of a machine in such a way that files in HDFS can be accessed as though they are stored locally. This requires providing the address of the NameNode machine, and the port on which the HDFS server listens for requests, to the local file system. The local file system recognizes which file accesses are to files in HDFS based on the file path, and sends appropriate requests to the HDFS server.

More details about distributed file system implementation may be found in Section 21.6.

10.2.2 Sharding

A single database system typically has sufficient storage and performance to handle all the transaction processing needs of an enterprise. However, using a single database is not sufficient for applications with millions or even billions of users, including social-media or similar web-scale applications, but also the user-facing applications of very large organizations such as large banks.

Suppose an organization has built an application with a centralized database, but needs to scale to handle more users, and the centralized database is not capable of handling the storage or processing speed requirements. A commonly used way to deal with such a situation is to partition the data across multiple databases, with a subset of users assigned to each of the databases. The term **sharding** refers to the partitioning of data across multiple databases or machines.

Partitioning is usually done on one or more attributes, referred to as *partitioning attributes*, *partitioning keys*, or *shard keys*. User or account identifiers are commonly used as partitioning keys. Partitioning can be done by defining a range of keys that each of the databases handles; for example, keys from 1 to 100,000 may be assigned to the

first database, keys from 100,001 to 200,000 to the second database, and so on. Such partitioning is called *range partitioning*. Partitioning may also be done by computing a hash function that maps a key value to a partition number; such partitioning is called *hash partitioning*. We study partitioning of data in detail in Chapter 21.

When sharding is done in application code, the application must keep track of which keys are stored on which database, and must route queries to the appropriate database. Queries that read or update data from multiple databases cannot be processed in a simple manner, since it is not possible to submit a single query that gets executed across all the databases. Instead, the application would need to read data from multiple databases and compute the final query result. Updates across databases cause further issues, which we discuss in Section 10.2.5.

While sharding performed by modifying application code provided a simple way to scale applications, the limitations of the approach soon became apparent. First, the application code has to track how data was partitioned and route queries appropriately. If a database becomes overloaded, parts of the data in that database have to be offloaded to a new database, or to one of the other existing databases; managing this process is a non-trivial task. As more databases are added, there is a greater chance of failure leading to loss of access to data. Replication is needed to ensure data is accessible despite failures, but managing the replicas, and ensuring they are consistent, poses further challenges. Key-value stores, which we study next, address some of these issues. Challenges related to consistency and availability are discussed later, in Section 10.2.5.

10.2.3 Key-Value Storage Systems

Many web applications need to store very large numbers (many billions or in extreme cases, trillions) of relatively small records (of size ranging from a few kilobytes to a few megabytes). Storing each record as a separate file is infeasible, since file systems, including distributed file systems, are not designed to store such large numbers of files.

Ideally, a massively parallel relational database should be used to store such data. However, it is not easy to build relational database systems that can run in parallel across a large number of machines while also supporting standard database features such as foreign-key constraints and transactions.

A number of storage systems have been developed that can scale to the needs of web applications and store large amounts of data, scaling to thousands to tens of thousands of machines, but typically offering only a simple key-value storage interface. A **key-value storage system** (or **key-value store**) is a system that provides a way to store or update a record (value) with an associated key and to retrieve the record with a given key.

Parallel key-value stores partition keys across multiple machines, and route updates and lookups to the correct machine. They also support replication, and ensure that replicas are kept consistent. Further, they provide the ability to add more machines to a system when required, and ensure that the load is automatically balanced across the machines in a system In contrast to systems that implement sharding in the application

code, systems that use a parallel key-value store do not need to worry about any of the above issues. Parallel key-value stores are therefore more widely used than sharding today.

Widely used parallel key-value stores include Bigtable from Google, Apache HBase, Dynamo from Amazon, Cassandra from Facebook, MongoDB, Azure cloud storage from Microsoft, and Sherpa/PNUTS from Yahoo!, among many others.

While several key-value data stores view the values stored in the data store as an uninterpreted sequence of bytes, and do not look at their content, other data stores allow some form of structure or schema to be associated with each record. Several such key-value storage systems require the stored data to follow a specified data representation, allowing the data store to interpret the stored values and execute simple queries based on stored values. Such data stores are called **document stores**. MongoDB is a widely used data store that accepts values in the JSON format.

Key-value storage systems are, at their core, based on two primitive functions, `put(key, value)`, used to store values with an associated key, and `get(key)`, used to retrieve the stored value associated with the specified key. Some systems, such as Bigtable, additionally provide range queries on key values. Document stores additionally support limited forms of querying on the data values.

An important motivation for the use of key-value stores is their ability to handle very large amounts of data as well as queries, by distributing the work across a *cluster* consisting of a large number of machines. Records are partitioned (divided up) among the machines in the cluster, with each machine storing a subset of the records and processing lookups and updates on those records.

Note that key-value stores are not full-fledged databases, since they do not provide many of the features that are viewed as standard on database systems today. Key-value stores typically do not support declarative querying (using SQL or any other declarative query language) and do not support transactions (which, as we shall see in Chapter 17, allow multiple updates to be committed atomically to ensure that the database state remains consistent despite failures, and control concurrent access to data to ensure that problems do not arise due to concurrent access by multiple transactions). Key-value stores also typically do not support retrieval of records based on selections on non-key attributes, although some document stores do support such retrieval.

An important reason for not supporting such features is that some of them are not easy to support on very large clusters; thus, most systems sacrifice these features in order to achieve scalability. Applications that need scalability may be willing to sacrifice these features in exchange for scalability.

Key-value stores are also called **NoSQL** systems, to emphasize that they do not support SQL, and the lack of support for SQL was initially viewed as something positive, rather than a limitation. However, it soon became clear that lack of database features such as transaction support and support for SQL, make application development more complicated. Thus, many key-value stores have evolved to support features, such as the SQL language and transactions.

```
show dbs // Shows available databases
use sampledb // Use database sampledb, creating it if it does not exist
db.createCollection("student") // Create a collection
db.createCollection("instructor")
show collections // Shows all collections in the database

db.student.insert({ "id" : "00128", "name" : "Zhang",
      "dept_name" : "Comp. Sci.", "tot_cred" : 102, "advisors" : ["45565"] })
db.student.insert({ "id" : "12345", "name" : "Shankar",
      "dept_name" : "Comp. Sci.", "tot_cred" : 32, "advisors" : ["45565"] })
db.student.insert({ "id" : "19991", "name" : "Brandt",
      "dept_name" : "History", "tot_cred" : 80, "advisors" : [] })
db.instructor.insert({ "id" : "45565", "name" : "Katz",
      "dept_name" : "Comp. Sci.", "salary" : 75000,
      "advisees" : ["00128","12345"] })

db.student.find() // Fetch all students in JSON format
db.student.findOne({"ID": "00128"}) // Find one matching student

db.student.remove({"dept_name": "Comp. Sci."}) // Delete matching students
db.student.drop() // Drops the entire collection
```

Figure 10.2 MongoDB shell commands.

The APIs provided by these systems to store and access data are widely used. While the basic get() and put() functions mentioned earlier are straightforward, most systems support further features. As an example of such APIs, we provide a brief overview of the MongoDB API.

Figure 10.2 illustrates access to the MongoDB document store through a JavaScript shell interface. Such a shell can be opened by executing the mongo command on a system that has MongoDB installed and configured. MongoDB also provides equivalent API functions in a variety of languages, including Java and Python. The use command shown in the figure opens the specified database, creating it if it does not already exist. The db.createCollection() command is used to create collections, which store *documents*; a document in MongoDB is basically a JSON object. The code in the figure creates two collections, student and instructor, and inserts JSON objects representing students and instructors into the two collections.

MongoDB automatically creates identifiers for the inserted objects, which can be used as keys to retrieve the objects. The key associated with an object can be fetched using the _id attribute, and an index on this attribute is created by default.

MongoDB also supports queries based on the stored values. The db.student.find() function returns a collection of all objects in the student collection, while the findOne() function returns one object from the collection. Both functions can take as argument a JSON object that specifies a selection on desired attributes. In our example, the

student with ID 00128 is retrieved. Similarly, all objects matching such a selection can be deleted by the remove() function shown in the figure. The drop() function shown in the figure drops an entire collection.

MongoDB supports a variety of other features such as creation of indices on specified attributes of the stored JSON objects, such as the ID and name attributes.

Since a key goal of MongoDB is to enable scaling to very large data sizes and query/update loads, MongoDB allows multiple machines to be part of a single MongoDB cluster. Data are then sharded (partitioned) across these machines. We study partitioning of data across machines in detail in Chapter 21, and we study parallel processing of queries in detail in Chapter 22. However we outline key ideas in this section.

In MongoDB (as in many other databases), partitioning is done based on the value of a specified attribute, called the **partitioning attribute** or **shard key**. For example, if we specify that the student collection should be partitioned on the dept_name attribute, all objects of a particular department are stored on one machine, but objects of different departments may be stored on different machines. To ensure data can be accessed even if a machine has failed, each partition is replicated on multiple machines. This way, even if one machine fails, the data in that partition can be fetched from another machine.

Requests from a MongoDB client are sent to a router, which then forwards requests to the appropriate partitions in a cluster.

Bigtable is another key-value store that requires data values to follow a format that allows the storage system access to individual parts of a stored value. In Bigtable, data values (records) can have multiple attributes; the set of attribute names is not predetermined and can vary across different records. Thus, the key for an attribute value conceptually consists of (record-identifier, attribute-name). Each attribute value is just a string as far as Bigtable is concerned. To fetch all attributes of a record, a range query, or more precisely a prefix-match query consisting of just the record identifier, is used. The get() function returns the attribute names along with the values. For efficient retrieval of all attributes of a record, the storage system stores entries sorted by the key, so all attribute values of a particular record are clustered together.

In fact, the record identifier can itself be structured hierarchically, although to Bigtable itself the record identifier is just a string. For example, an application that stores pages retrieved from a web crawl could map a URL of the form:

www.cs.yale.edu/people/silberschatz.html

to the record identifier:

edu.yale.cs.www/people/silberschatz.html

With this representation, all URLs of cs.yale.edu can be retried by a query that fetches all keys with the prefix edu.yale.cs, which would be stored in a consecutive range of key values in the sorted key order. Similarly, all URLs of yale.edu would have a prefix of edu.yale and would be stored in a consecutive range of key values.

Although Bigtable does not support JSON natively, JSON data can be mapped to the data model of Bigtable. For example, consider the following JSON data:

```
{   "ID": "22222",
    "name": { "firstname: "Albert", "lastname: "Einstein" },
    "deptname": "Physics",
    "children": [
        {"firstname": "Hans", "lastname": "Einstein" },
        {"firstname": "Eduard", "lastname": "Einstein" } ]
}
```

The above data can be represented by a Bigtable record with identifier "22222", with multiple attribute names such as "name.firstname", "deptname", "children[1].firstname" or "children[2].lastname".

Further, a single instance of Bigtable can store data for multiple applications, with multiple tables per application, by simply prefixing the application name and table name to the record identifier.

Many data-storage systems allow multiple versions of data items to be stored. Versions are often identified by timestamp, but they may be alternatively identified by an integer value that is incremented whenever a new version of a data item is created. Lookups can specify the required version of a data item or can pick the version with the highest version number. In Bigtable, for example, a key actually consists of three parts: (record-identifier, attribute-name, timestamp). Bigtable can be accessed as a service from Google. The open-source version of Bigtable, HBase, is widely used.

10.2.4 Parallel and Distributed Databases

Parallel databases are databases that run on multiple machines (together referred to as a cluster) and are designed to store data across multiple machines and to process large queries using multiple machines. Parallel databases were initially developed in the 1980s, and thus they predate the modern generation of Big Data systems. From a programmer viewpoint, parallel databases can be used just like databases running on a single machine.

Early generation parallel databases designed for transaction processing supported only a few machines in a cluster, while those designed to process large analytical queries were designed to support tens to hundreds of machines. Data are replicated across multiple machines in a cluster, to ensure that data are not lost, and they continue to be accessible, even if a machine in a cluster fails. Although failures do occur and need to be dealt with, failures during the processing of a query are not common in systems with tens to hundreds of machines. If a query was being processed on a node that failed, the query is simply restarted, using replicas of data that are on other nodes.

If such database systems are run on clusters with thousands of machines, the probability of failure during execution of a query increases significantly for queries that process a large amount of data and consequently run for a long time. Restarting a query in

the event of a failure is no longer an option, since there is a fairly high probability that a failure will happen yet again while the query is executing. Techniques to avoid complete restart, allowing only computation on the failed machines to be redone, were developed in the context of *map-reduce* systems, which we study in Section 10.3. However, these techniques introduce significant overhead; given the fact that computation spanning thousands to tens of thousands of nodes is needed only by some exceptionally large applications, even today most parallel relational database systems target applications that run on tens to hundreds of machines and just restart queries in the event of failure.

Query processing in such parallel and distributed databases is covered in detail in Chapter 22, while transaction processing in such databases is covered in Chapter 23.

10.2.5 Replication and Consistency

Replication is key to ensuring availability of data, ensuring a data item can be accessed despite failure of some of the machines storing the data item. Any update to a data item must be applied to all replicas of the data item. As long as all the machines containing the replicas are up and connected to each other, applying the update to all replicas is straightforward.

However, since machines do fail, there are two key problems. The first is how to ensure atomic execution of a transaction that updates data at more than one machine: the transaction execution is said to be atomic if despite failures, either all the data items updated by the transaction are successfully updated, or all the data items are reverted back to their original values. The second problem is, how to perform updates on a data item that has been replicated, when some of the replicas of the data item are on a machine that has failed. A key requirement here is *consistency*, that is, all live replicas of a data item have the same value, and each read sees the latest version of the data item. There are several possible solutions, which offer different degrees of resilience to failures. We study solutions to the both these problems in Chapter 23.

We note that the solutions to the second problem typically require that a majority of the replicas are available for reading and update. If we had 3 replicas, this would require not more than 1 fail, but if we had 5 replicas, even if two machines fail we would still have a majority of replicas available. Under these assumptions, writes will not get blocked, and reads will see the latest value for any data item.

While the probability of multiple machines failing is relatively low, network link failures can cause further problems. In particular, a *network partition* is said to occur if two live machines in a network are unable to communicate with each other.

It has been shown that no protocol can ensure *availability*, that is, the ability to read and write data, while also guaranteeing consistency, in the presence of network partitions. Thus, distributed systems need to make tradeoffs: if they want high availability, they need to sacrifice consistency, for example by allowing reads to see old values of data items, or to allow different replicas to have different values. In the latter case, how to bring the replicas to a common value by merging the updates is a task that the

Note 10.1 Building Scalable Database Applications

When faced with the task of creating a database application that can scale to a very large number of users, application developers typically have to choose between a database system that runs on a single server, and a key-value store that can scale by running on a large number of servers. A database that supports SQL and atomic transactions, and at the same time is highly scalable, would be ideal; as of 2018, Google Cloud Spanner, which is only available on the cloud, and the recently developed open source database CockroachDB are the only such databases.

Simple applications can be written using only key-value stores, but more complex applications benefit greatly from having SQL support. Application developers therefore typically use a combination of parallel key-value stores and databases.

Some relations, such as those that store user account and user profile data are queried frequently, but with simple select queries on a key, typically on the user identifier. Such relations are stored in a parallel key-value store. In case select queries on other attributes are required, key-value stores that support indexing on attributes other than the primary key, such as MongoDB, could still be used.

Other relations that are used in more complex queries are stored in a relational database that runs on a single server. Databases running on a single server do exploit the availability of multiple cores to execute transactions in parallel, but are limited by the number of cores that can be supported in a single machine.

Most relational databases support a form of replication where update transactions run on only one database (the primary), but the updates are propagated to replicas of the database running on other servers. Applications can execute read-only queries on these replicas, but with the understanding that they may see data that is a few seconds behind in time, as compared to the primary database. Offloading read-only queries from the primary database allows the system to handle a load larger than what a single database server can handle.

In-memory caching systems, such as memcached or Redis, are also used to get scalable read-only access to relations stored in a database. Applications may store some relations, or some parts of some relations, in such an in-memory cache, which may be replicated or partitioned across multiple machines. Thereby, applications can get fast and scalable read-only access to the cached data. Updates must however be performed on the database, and the application is responsible for updating the cache whenever the data is updated on the database.

application has to deal with. Some applications, or some parts of an application, may choose to prioritize availability over consistency. But other applications, or some parts of an application, may choose to prioritize consistency even at the cost of potential non-availability of the system in the event of failures. The above issues are discussed in more detail in Chapter 23.

10.3 The MapReduce Paradigm

The MapReduce paradigm models a common situation in parallel processing, where some processing, identified by the map() function, is applied to each of a large number of input records, and then some form of aggregation, identified by the reduce() function, is applied to the result of the map() function. The map() function is also permitted to specify grouping keys, such that the aggregation specified in the reduce() function is applied within each group, identified by the grouping key, of the map() output. We examine the MapReduce paradigm, and the map() and reduce() functions in detail, in the rest of this section.

The MapReduce paradigm for parallel processing has a long history, dating back several decades, in the functional programming and parallel processing community (the map and reduce functions were supported in the Lisp language, for example).

10.3.1 Why MapReduce?

As a motivating example for the use of the MapReduce paradigm, we consider the following *word count* application, which takes a large number of files as input, and outputs a count of the number of times each word appears, across all the files. Here, the input would be in the form of a potentially large number of files stored in a directory.

We start by considering the case of a single file. In this case, it is straightforward to write a program that reads in the words in the file and maintains an in-memory data structure that keeps track of all the words encountered so far, along with their counts. The question is, how to extend the above algorithm, which is sequential in nature, to an environment where there are tens of thousands of files, each containing tens to hundreds of megabytes of data. It is infeasible to process such a large volume of data sequentially.

One solution is to extend the above scheme by coding it as a parallel program that would run across many machines with each machine processing a part of the files. The counts computed locally at each machine must then be combined to get the final counts. In this case, the programmer would be responsible for all the "plumbing" required to start up jobs on different machines, coordinate them, and to compute the final answer. In addition, the "plumbing" code must also deal with ensuring completion of the program in spite of machine failures; failures are quite frequent when the number of participating machines is large, such as in the thousands, and the program runs for a long duration.

The "plumbing" code to implement the above requirements is quite complex; it makes sense to write it just once and reuse it for all desired applications.

MapReduce systems provide the programmer a way of specifying the core logic needed for an application, with the details of the earlier-mentioned plumbing handled by the MapReduce system. The programmer needs to provide only map() and reduce() functions, plus optionally functions for reading and writing data. The map() and reduce() functions provided by the programmer are invoked on the data by the MapRe-

duce system to process data in parallel. The programmer does not need to be aware of the plumbing or its complexity; in fact, she can for the most part ignore the fact that the program is to be executed in parallel on multiple machines.

The MapReduce approach can be used to process large amounts of data for a variety of applications. The above-mentioned word count program is a toy example of a class of text and document processing applications. Consider, for example, search engines which take keywords and return documents containing the keywords. MapReduce can, for example, be used to process documents and create text indices, which are then used to efficiently find documents containing specified keywords.

10.3.2 MapReduce By Example 1: Word Count

Our word count application can be implemented in the MapReduce framework using the following functions, which we defined in pseudocode. Note that our pseudocode is not in any specific programming language; it is intended to introduce concepts. We describe how to write MapReduce code in specific languages in later sections.

1. In the MapReduce paradigm, the `map()` function provided by the programmer is invoked on each input record and emits zero or more output data items, which are then passed on to the `reduce()` function. The first question is, what is a record? MapReduce systems provide defaults, treating each line of each input file as a record; such a default works well for our word count application, but the programmers are allowed to specify their own functions to break up input files into records.

   For the word count application, the `map()` function could break up each record (line) into individual words and output a number of records, each of which is a pair (*word, count*), where count is the number of occurrences of the word in the record. In fact in our simplified implementation, the `map()` function does even less work and outputs each word as it is found, with a count of 1. These counts are added up later by the `reduce()`. Pseudocode for the `map()` function for the word count program is shown in Figure 10.3.

   The function breaks up the record (line) into individual words.[2] As each word is found, the `map()` function emits (outputs) a record (*word*, 1). Thus, if the file contained just the sentence:

   ```
   "One a penny, two a penny, hot cross buns."
   ```

   the records output by the `map()` function would be

   ```
   ("one", 1), ("a", 1), ("penny", 1),("two", 1), ("a", 1), ("penny", 1),
      ("hot", 1), ("cross", 1), ("buns", 1).
   ```

[2] We omit details of how a line is broken up into words. In a real implementation, non-alphabet characters would be removed, and uppercase characters mapped to lowercase, before breaking up the line based on spaces to generate a list of words.

```
map(String record) {
      For each word in record
            emit(word, 1).
}

reduce(String key, List value_list) {
      String word = key;
      int count = 0;
      For each value in value_list
            count = count + value
      output(word, count)
}
```

Figure 10.3 Pseudocode of map-reduce job for word counting in a set of files.

In general, the map() function outputs a set of (*key, value*) pairs for each input record. The first attribute (key) of the map() output record is referred to as a **reduce key**, since it is used by the reduce step, which we study next.

2. The MapReduce system takes all the (*key, value*) pairs emitted by the map() functions and sorts (or at least, groups them) such that all records with a particular key are gathered together. All records whose keys match are grouped together, and a list of all the associated values is created. The (*key, list*) pairs are then passed to the reduce() function.

   In our word count example, each key is a word, and the associated list is a list of counts generated for different lines of different files. With our example data, the result of this step is the following:

   ("a", [1,1]), ("buns", [1]) ("cross", [1]), ("hot", [1]), ("one", [1]), ("penny", [1,1]), ("two", [1])

   The reduce() function for our example combines the list of word counts by adding the counts, and outputs (*word, total-count*) pairs. For the example input, the records output by the reduce() function would be as follows:

   ("one", 1), ("a", 2), ("penny", 2), ("two", 1), ("hot", 1), ("cross", 1), ("buns", 1).

   Pseudocode for the reduce() function for the word count program is shown in Figure 10.3. The counts generated by the map() function are all 1, so the reduce() function could have just counted the number of values in the list, but adding up the values allows some optimizations that we will see later.

   A key issue here is that with many files, there may be many occurrences of the same word across different files. Reorganizing the outputs of the map() functions

```
...
2013/02/21 10:31:22.00EST /slide-dir/11.ppt
2013/02/21 10:43:12.00EST /slide-dir/12.ppt
2013/02/22 18:26:45.00EST /slide-dir/13.ppt
2013/02/22 18:26:48.00EST /exer-dir/2.pdf
2013/02/22 18:26:54.00EST /exer-dir/3.pdf
2013/02/22 20:53:29.00EST /slide-dir/12.ppt
...
```

Figure 10.4 Log files.

is required to bring all the values for a particular key together. In a parallel system with many machines, this requires data for different reduce keys to be exchanged between machines, so all the values for any particular reduce key are available at a single machine. This work is done by the **shuffle step**, which performs data exchange between machines and then sorts the (*key, value*) pairs to bring all the values for a key together. Observe in our example that the words have actually been sorted alphabetically. Sorting the output records from the map() is one way for the system to collect all occurrences of a word together; the lists for each word are created from the sorted records.

By default, the output of the reduce() function is sent to one or more files, but MapReduce systems allow programmers to control what happens to the output.

10.3.3 MapReduce by Example 2: Log Processing

As another example of the use of the MapReduce paradigm, which is closer to traditional database query processing, suppose we have a log file recording accesses to a web site, which is structured as shown in Figure 10.4. The goal of our file access count application is to find how many times each of the files in the slide-dir directory was accessed between 2013/01/01 and 2013/01/31. The application illustrates one of a variety of kinds of questions an analyst may ask using data from web log files.

For our log-file processing application, each line of the input file can be treated as a record. The map() function would do the following: it would first break up the input record into individual fields, namely date, time, and filename. If the date is in the required date range, the map() function would emit a record (*filename, 1*), which indicates that the filename appeared once in that record. Pseudocode for the map() function for this example is shown in Figure 10.5.

The shuffle step brings all the values for a particular *reduce* key (in our case, a file name) together as a list. The reduce() function provided by the programmer, shown in Figure 10.6, is then invoked for each *reduce* key value. The first argument of reduce() is the *reduce* key itself, while the second argument is a list containing the values in the

```
map(String record) {
    String attribute[3];
    break up record into tokens (based on space character), and
        store the tokens in array attributes
    String date = attribute[0];
    String time = attribute[1];
    String filename = attribute[2];
    if(date between 2013/01/01 and 2013/01/31
            and filename starts with "http://db-book.com/slide-dir")
        emit(filename, 1).
}
```

Figure 10.5 Pseudocode of map functions for counting file accesses.

records emitted by the map() function for that *reduce* key. In our example, the values for a particular key are added to get the total number of accesses for a file. This number is then output by the reduce() function.

If we were to use the values generated by the map() function, the values would be "1" for all emitted records, and we could have just counted the number of elements in the list. However, MapReduce systems support optimizations such as performing a partial addition of values from each input file, before they are redistributed. In that case, the values received by the reduce() function may not necessarily be ones, and we therefore add the values.

Figure 10.7 shows a schematic view of the flow of keys and values through the map() and reduce() functions. In the figure the mk_i's denote *map* keys, mv_i's denote map input values, rk_i's denote *reduce* keys, and rv_i's denote reduce input values. Reduce outputs are not shown.

```
reduce(String key, List value_list) {
    String filename = key;
    int count = 0;
    For each value in value_list
        count = count + value
    output(filename, count)
}
```

Figure 10.6 Pseudocode of reduce functions for counting file accesses.

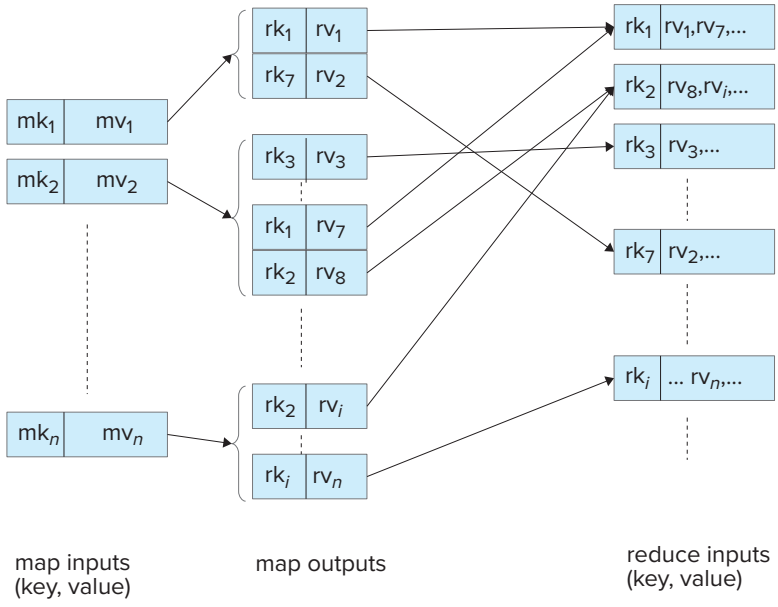


Figure 10.7 Flow of keys and values in a MapReduce job.

10.3.4 Parallel Processing of MapReduce Tasks

Our description of the `map()` and `reduce()` functions so far has ignored the issue of parallel processing. We can understand the meaning of MapReduce code without considering parallel processing. However, our goal in using the MapReduce paradigm is to enable parallel processing. Thus, MapReduce systems execute the `map()` function in parallel on multiple machines, with each map task processing some part of the data, for example some of the files, or even parts of a file in case the input files are very large. Similarly, the `reduce()` functions are also executed in parallel on multiple machines, with each reduce task processing a subset of the reduce keys (note that a particular call to the `reduce()` function is still for a single reduce key).

Parallel execution of `map` and `reduce` tasks is shown pictorially in Figure 10.8. In the figure, the input file partitions, denoted as Part *i*, could be files or parts of files. The nodes denoted as Map *i* are the map tasks, and the nodes denoted Reduce *i* are the reduce tasks. The master node sends copies of the `map()` and `reduce()` code to the map and reduce tasks. The map tasks execute the code and write output data to local files on the machines where the tasks are executed, after being sorted and partitioned based on the reduce key values; separate files are created for each reduce task at each Map node. These files are fetched across the network by the reduce tasks; the files fetched by a reduce task (from different map tasks) are merged and sorted to ensure that all occurrences of a particular reduce key are together in the sorted file. The reduce keys and values are then fed to the `reduce()` functions.

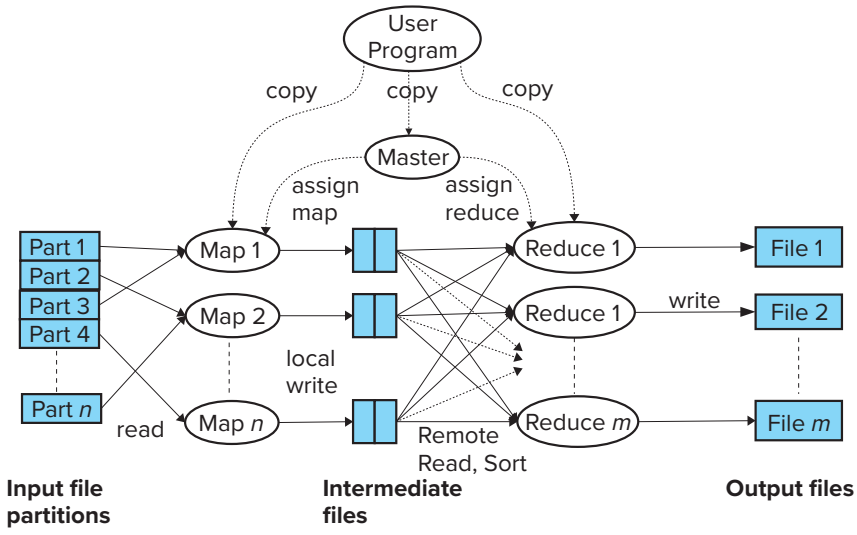


Figure 10.8 Parallel processing of MapReduce job.

MapReduce systems also need to parallelize file input and output across multiple machines; otherwise the single machine storing the data will become a bottleneck. Parallelization of file input and output can be done by using a distributed file system, such as the *Hadoop File System* (*HDFS*). As we saw in Section 10.2, distributed file systems allow a number of machines to cooperate in storing files, partitioning the files across the machines. Further, file system data are replicated (copied) across several (typically three) machines, so that even if a few of the machines fail, the data are available from other machines which have copies of the data in the failed machine.

Today, in addition to distributed file systems such as HDFS, MapReduce systems support input from a variety of Big Data storage systems such as HBase, MongoDB, Cassandra, and Amazon Dynamo, by using storage adapters. Output can similarly be sent to any of these storage systems.

10.3.5 MapReduce in Hadoop

The Hadoop project provides a widely used open-source implementation of MapReduce in the Java language. We summarize its main features here using the Java API provided by Hadoop. We note that Hadoop provides MapReduce APIs in several other languages, such as Python and C++.

Unlike our MapReduce pseudocode, real implementations such as Hadoop require types to be specified for the input keys and values, as well as the output keys and value, of the `map()` function. Similarly, the types of the input as well as output keys and values of the `reduce()` function need to be specified. Hadoop requires the programmer to implement `map()` and `reduce()` functions as member functions of classes that extend Hadoop `Mapper` and `Reducer` classes. Hadoop allows the programmer to provide

functions to break up the file into records, or to specify that the file is one of the file types for which Hadoop provides built-in functions to break up files into records. For example, the TextInputFormat specifies that the file should be broken up into lines, with each line being a separate record. Compressed file formats are widely used today, with *Avro*, *ORC,* and *Parquet* being the most widely used compressed file formats in the Hadoop world (compressed file formats are discussed in Section 13.6). Decompression is done by the system, and a programmer writing a query need only specify one of the supported types, and the uncompressed representation is made available to the code implementing the query.

Input files in Hadoop can come from a file system of a single machine, but for large datasets, a file system on a single machine would become a performance bottleneck. Hadoop MapReduce allows input and output files to be stored in a distributed file system such as HDFS, allowing multiple machines to read and write data in parallel.

In addition to the reduce() function, Hadoop also allows the programmer to define a combine() function, which can perform a part of the reduce() operation at the node where the map() function is executed. In our word count example, the combine() function would be the same as the reduce() function we saw earlier. The reduce() function would then receive a list of partial counts for a particular word; since the reduce() function for word count adds up the values, it would work correctly even with the combine() function. One benefit of using the combine() function is that it reduces the amount of data that has to be sent over the network: each node that runs map tasks would send only one entry for a word across the network, instead of multiple entries.

A single MapReduce step in Hadoop executes a map and a reduce function. A program may have multiple MapReduce steps, with each step having its own map and reduce functions. The Hadoop API allows a program to execute multiple such MapReduce steps. The reduce() output from each step is written to the (distributed) file system and read back in the following step. Hadoop also allows the programmer to control the number of map and reduce tasks to be run in parallel for the job.

The rest of this section assumes a basic knowledge of Java (you may skip the rest of this section without loss of continuity, if you are not familiar with Java).

Figure 10.9 shows the Java implementation in Hadoop of the word count application we saw earlier. For brevity we have omitted Java import statements. The code defines two classes, one that implements the Mapper interface, and another that implements the Reducer interface. The Mapper and Reducer classes are generic classes which take as arguments the types of the keys and values. Specifically, the generic Mapper and Reducer interfaces both takes four type arguments that specify the types of the input key, input value, output key, and output value, respectively.

The type definition of the Map class in Figure 10.9, which implements the Mapper interface, specifies that the map key is of type LongWritable, is basically a long integer, and the value which is (all or part of) a document is of type Text. The output of map has a key of type Text, since the key is a word, while the value is of type IntWritable, which is an integer value.

```
public class WordCount {
    public static class Map extends Mapper<LongWritable, Text, Text, IntWritable> {
        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(LongWritable key, Text value, Context context)
                                    throws IOException, InterruptedException {
            String line = value.toString();
            StringTokenizer tokenizer = new StringTokenizer(line);
            while (tokenizer.hasMoreTokens()) {
                word.set(tokenizer.nextToken());
                context.write(word, one);
            }
        }
    }
    public static class Reduce extends Reducer<Text, IntWritable, Text, IntWritable> {
        public void reduce(Text key, Iterable<IntWritable> values, Context context)
                                    throws IOException, InterruptedException {
            int sum = 0;
            for (IntWritable val : values) {
                sum += val.get();
            }
            context.write(key, new IntWritable(sum));
        }
    }
    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = new Job(conf, "wordcount");

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        job.setMapperClass(Map.class);
        job.setReducerClass(Reduce.class);
        job.setInputFormatClass(TextInputFormat.class);
        job.setOutputFormatClass(TextOutputFormat.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        job.waitForCompletion(true);
    }
}
```

Figure 10.9 The word count program written in Hadoop.

The map() code for the word count example breaks up the input text value into words using StringTokenizer, and then for each word, it invokes context.write(word,

one) to output a key and value pair; note that one is an IntWritable object with numeric value 1.

All the values output by the map() invocations that have a particular key (word, in our example) are collected in a list by the MapReduce system infrastructure. Doing so requires interchange of data from multiple map tasks to multiple reduce tasks; in a distributed setting, the data would have to be sent over the network. To ensure that all values for a particular key come together, the MapReduce system typically sorts the keys output by the map functions, ensuring all values for a particular key will come together in the sorted order. This list of values for each key is provided to the reduce() function.

The type of the reduce() input key is the same as the type of the map output key. The reduce() input value in our example is a Java Iterable<IntWritable> object, which contains a list of map output values (IntWritable is the type of the map output value). The output key for reduce() is a word, of type Text, while the output value is a word count, of type IntWritable.

In our example, the reduce() simply adds up the values it receives in its input to get the total count; reduce() writes the word and the total count using the context.write() function.

Note that in our simple example, the values are all 1, so reduce() just needs to count the number of values it receives. In general, however, Hadoop allows the programmer to declare a Combiner class, whose combine() function is run on the output of a single map job; the output of this function replaces multiple map() output values for a single key with a single value. In our example, a combine() function could just count the number of occurrences of each word and output a single value, which is the local word count at the map task. These outputs are then passed on to the reduce() function, which would add up the local counts to get the overall count. The Combiner's job is to reduce the traffic over the network.

A MapReduce job runs a map and a reduce step. A program may have multiple MapReduce steps, and each step would have its own settings for the map and reduce functions. The main() function sets up the parameters for each MapReduce job, and then executes it.

The example code in Figure 10.9 executes a single MapReduce job; the parameters for the job are as follows:

- The classes that contain the map and reduce functions for the job, set by the methods setMapperClass and setReducerClass.
- The types of the job's output key and values, set to Text (for the words) and IntWritable (for the count), respectively, by methods setOutputKeyClass and setOutputValueClass, respectively.
- The input format of the job, set to TextInputFormat by the method job.setInputFormatClass. The default input format in Hadoop is the TextInputFormat, which creates a map key whose value is a byte offset into the file, and the

map value is the contents of one line of the file. Since files are allowed to be bigger than 4 gigabytes, the offset is of type LongWritable. Programmers can provide their own implementations for the input format class, which would process input files and break the files into records.

- The output format of the job, set to TextOutputFormat by the method job.setOutputFormatClass.
- The directories where the input files are stored, and where the output files must be created, set by the methods addInputPath and addOutputPath.

Hadoop supports many more parameters for MapReduce jobs, such as the number of map and reduce tasks to be run in parallel for the job and the amount of memory to be allocated to each map and reduce task, among many others.

10.3.6 SQL on MapReduce

Many of the applications of MapReduce are for parallel processing of large amounts of non-relational data, using computations that cannot be expressed easily in SQL. For example, our word count program cannot be expressed easily in SQL. There are many real-world uses of MapReduce that cannot be expressed in SQL. Examples include computation of "inverted indices" which are key for web search engines to efficiently answer keyword queries, and computation of Google's PageRank, which is an important measure of the importance of web sites, and is used to rank answers to web search queries.

However, there are a large number of applications that have used the MapReduce paradigm for data processing of various kinds, whose logic can be easily expressed using SQL. If the data were in a database, it would make sense to write such queries using SQL and execute the queries on a parallel database system (parallel database systems are discussed in detail in Chapter 22. Using SQL is much easier for users than is coding in the MapReduce paradigm. However, the data for many such applications reside in a file system, and there are significant time and space overhead demands when loading them into a database.

Relational operations can be implemented using map and reduce steps, as illustrated by the following examples:

- The relational selection operation can be implemented by a single map() function, without a reduce() function (or with a reduce() function that simply outputs its inputs, without any change).
- The relational group by and aggregate function γ can be implemented using a single MapReduce step: the map() outputs records with the group by attribute values as the reduce key; the reduce() function receives a list of all the attribute values for a particular group by key and computes the required aggregate on the values in its input list.

- A join operation can be implemented using a single MapReduce step, Consider the equijoin operation $r \bowtie_{r.A=s.A} s$. We define a map() function which for each input record r_i outputs a pair $(r_i.A, r_i)$, and similarly for each input record s_i outputs a pair $(s_i.A, s_i)$; the map output also includes a tag to indicate which relation (r or s) the output came from. The reduce() function is invoked for each join-attribute value, with a list of all the r_i and s_i records with that join-attribute value. The function separates out the r and s tuples, and then outputs a cross products of the r tuples and the s tuples, since all of them have the same value for the join attribute.

We leave details as an exercise to the reader (Exercise 10.4). More complex tasks, for example a query with multiple operations, can be expressed using multiple stages of map and reduce tasks.

While it is indeed possible for relational queries to be expressed using the MapReduce paradigm, it can be very cumbersome for a human to do so. Writing queries in SQL is much more concise and easy to understand, but traditional databases did not allow data access from files, nor did they support parallel processing of such queries.

A new generation of systems have been developed that allows queries written in (variants of) the SQL language to be executed in parallel on data stored in file systems. These systems include *Apache Hive* (which was initially developed at Facebook), *SCOPE*, which developed by Microsoft, both of which use variants of SQL, and *Apache Pig* (which was initially developed at Yahoo!), which uses a declarative language called *Pig Latin*, based on the relational algebra. All these systems allow data to be read directly from the file system but allow the programmer to define functions that convert the input data to a record format.

All these systems generate a program containing a sequence of map and reduce tasks to execute a given query. The programs are compiled and executed on a MapReduce framework such as Hadoop. These systems became very popular, and far more queries are written using these systems today than are written directly using the MapReduce paradigm.

Today, Hive implementations provide an option of compiling SQL code to a tree of algebraic operations that are executed on a parallel environment. Apache Tez and Spark are two widely used platforms that support the execution of a tree (or DAG) of algebraic operations on a parallel environment, which we study next in Section 10.4.

10.4 Beyond MapReduce: Algebraic Operations

Relational algebra forms the foundation of relational query processing, allowing queries to be modeled as trees of operations. This idea is extended to settings with more complex data types by supporting algebraic operators that can work on datasets containing records with complex data types, and returning datasets with records containing similar complex data types.

10.4.1 Motivation for Algebraic Operations

As we saw in Section 10.3.6, relational operations can be expressed by a sequence of map and reduce steps. Expressing tasks in such as fashion can be quite cumbersome.

For example, if programmers need to compute the join of two inputs, they should be able to express it as a single algebraic operation, instead of having to express it indirectly via map and reduce functions. Having access to functions such as joins can greatly simplify the job of a programmer.

The join operation can be executed in parallel, using a variety of techniques that we will see later in Section 22.3. In fact, doing so can be much more efficient than implementing the join using map and reduce functions. Thus, even systems like Hive, where programmers do not directly write MapReduce code, can benefit from direct support for operations such as join.

Later-generation parallel data-processing systems therefore added support for other relational operations such as joins (including variants such as outerjoins and semi-joins), as well as a variety of other operations to support data analytics. For example, many machine-learning models can be modeled as operators that take a set of records as input then output a set of records that have an extra attribute containing the value predicted by the model based on the other attributes of the record. Machine-learning algorithms can themselves be modeled as operators that take a set of training records as input and output a learned model. Processing of data often involves multiple steps, which can be modeled as a sequence (pipeline) or tree of operators.

A unifying framework for these operations is to treat them as *algebraic operations* that take one or more datasets as inputs and output one or more datasets.

Recall that in the relational algebra (Section 2.6) each operation takes one or more relations as input, and outputs a relation. These later-generation parallel query-processing systems are based on the same idea, but there are several differences. A key difference is that the input data could be of arbitrary types, instead of just consisting of columns with atomic data types as in the relational model. Recall that the extended relational algebra required to support SQL could restrict itself to simple arithmetic, string, and boolean expressions. In contrast, the new-generation algebraic operators need to support more complex expressions, requiring the full power of a programming language.

There are a number of frameworks that support algebraic operations on complex data; the most widely used ones today are Apache Tez and Apache Spark.

Apache Tez provides a low-level API which is suitable for system implementors. For example, Hive on Tez compiles SQL queries into algebraic operations that run on Tez. Tez programmers can create trees (or in general Directed Acyclic Graphs, or DAGs) of nodes, and they provide code that is to be executed on each of the nodes. Input nodes would read in data from data sources and pass them to other nodes, which operate on the data. Data can be partitioned across multiple machines, and the code for each node can be executed on each of the machines. Since Tez is not really designed for application programmers to use directly, we do not describe it in further detail.

However, Apache Spark provides higher-level APIs which are suitable for application programmers. We describe Spark in more detail next.

10.4.2 Algebraic Operations in Spark

Apache Spark is a widely used parallel data processing system that supports a variety of algebraic operations. Data can be input from or output to a variety of storage systems.

Just as relational databases use a relation as the primary abstraction for data representation, Spark uses a representation called a **Resilient Distributed Dataset** (**RDD**), which is a collection of records that can be stored across multiple machines. The term *distributed* refers to the records being stored on different machines, and *resilient* refers to the resilience to failure, in that even if one of the machines fails, records can be retrieved from other machines where they are stored.

Operators in Spark take one or more RDDs as input, and their output is an RDD. The types of records stored in RDDs is not predefined and can be anything that the application desires. Spark also supports a relational data representation called a DataSet, which we describe later.

Spark provides APIs for Java, Scala, and Python. Our coverage of Spark is based on the Java API.

Figure 10.10 shows our word count application, written in Java using Apache Spark; this program uses the RDD data representation, whose Java type is called JavaRDD. Note that JavaRDDs require a type for the record, specified in angular brackets ("<>"). In the program we have RDDs of Java Strings. The program also has JavaPairRDD types, which store records with two attributes of specified types. Records with multiple attributes can be represented by using structured data types instead of primitive data types. While any user-defined data type can be used, the predefined data types Tuple2 which stores two attributes, Tuple3, which stores three attributes, and Tuple4, which stores four attributes, are widely used.

The first step in processing data using Spark is to convert data from input representation to the RDD representation, which is done by the spark.read().textfile() function, which creates a record for each line in the input. Note that the input can be a file or a directory with multiple files; a Spark system running on multiple nodes will actually partition the RDD across multiple machines, although the program can treat it (for most purposes) as if it is a data structure on a single machine. In our sample code in Figure 10.10, the result is the RDD called lines.

The next step in our Spark program is to split each line into an array of words, by calling s.split(" ")) on the line; this function breaks up the line based on spaces and returns an array of words; a more complete function would split the input on other punctuation characters such as periods, semicolons, and so on. The split function can be invoked on each line in the input RDD by calling the map() function, which in Spark returns a single record for each input record. In our example, we instead use a variant called flatMap(), which works as follows: like map(), flatMap() invokes a user-defined

```
import java.util.Arrays;
import java.util.List;
import scala.Tuple2;
import org.apache.spark.api.java.JavaPairRDD;
import org.apache.spark.api.java.JavaRDD;
import org.apache.spark.sql.SparkSession;
public class WordCount {

    public static void main(String[] args) throws Exception {

        if (args.length < 1) {
            System.err.println("Usage: WordCount <file-or-directory-name>");
            System.exit(1);
        }
        SparkSession spark =
            SparkSession.builder().appName("WordCount").getOrCreate();

        JavaRDD<String> lines = spark.read().textFile(args[0]).javaRDD();
        JavaRDD<String> words = lines.flatMap(s -> Arrays.asList(s.split(" ")).iterator());
        JavaPairRDD<String, Integer> ones = words.mapToPair(s -> new Tuple2<>(s, 1));
        JavaPairRDD<String, Integer> counts = ones.reduceByKey((i1, i2) -> i1 + i2);

        counts.saveAsTextFile("outputDir"); // Save output files in this directory

        List<Tuple2<String, Integer>> output = counts.collect();
        for (Tuple2<String,Integer> tuple : output) {
            System.out.println(tuple);
        }
        spark.stop();
    }
}
```

Figure 10.10 Word count program in Spark.

function on each input record; the function is expected to return an iterator. A Java iterator supports a `next()` function that can be used to fetch multiple results by calling the function multiple times. The `flatMap()` function invokes the user-defined function to get an iterator, invokes the `next()` function repeatedly on the iterator to get multiple values, and then returns an RDD containing the union of all the values across all input records.

The code shown in Figure 10.10 uses the "lambda expression" syntax introduced in Java 8, which allows functions to be defined compactly, without even giving them a name; in the Java code, the syntax

```
s -> Arrays.asList(s.split(" ")).iterator()
```

defines a function that takes a parameter s and returns an expression that does the following: it applies the split function described earlier to create an array of words, then uses Arrays.asList to convert the array to a list, and finally applies the iterator() method on the list to create an iterator. The flatMap() function works on this iterator as described earlier.

The result of the above steps is an RDD called words, where each record contains a single word.

The next step is to create a JavaPairRDD called ones, which contains pairs of the form "(word, 1)" for each word in words; if a word appears multiple times in the input file, there would correspondingly be as many records in words and in ones.

Finally the algebraic operation reduceByKey() implements a group by and aggregation step. In the sample code, we specify that addition is to be used for aggregation, by passing the lambda function (i1, i2) -> i1+i2 to the reduceByKey() function. The reduceByKey() function works on a JavaPairRDD, grouping by the first attribute, and aggregating the values of the second attribute using the provided lambda function. When applied on the ones RDD, grouping would be on the word, which is the first attribute, and the values of the second attribute (all ones, in the ones RDD) would be added up. The result is stored in the JavaPairRDD counts.

In general, any binary function can be used to perform the aggregation, as long as it gives the same result regardless of the order in which it is applied on a collection of values.

Finally, the counts RDD is stored to the file system by saveAsTextFile(). Instead of creating just one file, the function creates multiple files if the RDD itself is partitioned across machines.

Key to understanding how parallel processing is achieved is to understand that

- RDDs may be partitioned and stored on multiple machines, and
- each operation may be executed in parallel on multiple machines, on the RDD partition available at the machine. Operations may first repartition their input, to bring related records to the same machine before executing operations in parallel. For example, reduceByKey() would repartition the input RDD to bring all records belonging to a group together on a single machine; records of different groups may be on different machines.

Another important aspect of Spark is that the algebraic operations are not necessarily evaluated immediately on the function call, although the code seems to imply that this is what happens. Instead, the code shown in the figure actually creates a tree of operations; in our code, the leaf operation textFile() reads data from a file; the next operation flatMap() has the textFile() operation as its child; the mapToPairs() in turn has flatMap() as child, and so on. The operators can be thought of in relational terms as defining views, which are not executed as soon as they are defined but get executed later.

The entire tree of operations actually get evaluated only when certain operations demand that the tree be evaluated. For example, saveAsTextFile() forces the tree to be evaluated; other such functions include collect(), which evaluates the tree and brings all records to a single machine, where they can subsequently be processed, for example by printing them out.

An important benefit of such *lazy evaluation* of the tree (i.e., the tree is evaluated when required, rather than when it is defined) is that before actual evaluation, it is possible for a query optimizer to rewrite the tree to another tree that computes the same result but may execute faster. Query optimization techniques, which we study in Chapter 16 can be applied to optimize such trees.

While the preceding example created a tree of operations, in general the operations may form a *Directed Acyclic Graph* (*DAG*) structure, if the result of an operation is consumed by more than one other operation. That would result in operations having more than one parent, leading to a DAG structure, whereas operations in a tree can have at most one parent.

While RDDs are well suited for representing certain data types such as textual data, a very large fraction of Big Data applications need to deal with structured data, where each record may have multiple attributes. Spark therefore introduced the DataSet type, which supports records with attributes. The DataSet type works well with widely used Parquet, ORC, and Avro file formats (discussed in more detail later in Section 13.6), which are designed to store records with multiple attributes in a compressed fashion. Spark also supports JDBC connectors that can read relations from a database.

The following code illustrates how data in Parquet format can be read and processed in Spark, where spark is a Spark session that has been opened earlier.

```
Dataset<Row> instructor = spark.read().parquet("...");
Dataset<Row> department = spark.read().parquet("...");
instructor.filter(instructor.col("salary").gt(100000))
      .join(department, instructor.col("dept_name")
      .equalTo(department.col("dept_name")))
      .groupBy(department.col("building"))
      .agg(count(instructor.col("ID")));
```

The DataSet<Row> type above uses the type Row, which allows access to column values by name. The code reads *instructor* and *department* relations from Parquet files (whose names are omitted in the code above); Parquet files store metadata such as column names in addition to the values, which allows Spark to create a schema for the relations. The Spark code then applies a filter (selection) operation on the *instructor* relation, which retains only instructors with salary greater than 100000, then joins the result with the *department* relation on the *dept_name* attribute, performs a group by on the *building* attribute (an attribute of the department relation), and for each group (here, each building), a count of the number of *ID* values is computed.

The ability to define new algebraic operations and to use them in queries has been found to be very useful for many applications and has led to wide adoption of Spark. The Spark system also supports compilation of Hive SQL queries into Spark operation trees, which are then executed.

Spark also allows classes other than Row to be used with DataSets. Spark requires that for each attribute *Attrk* of the class, methods *getAttrk*() and *setAttrk*() must be defined to allow retrieval and storage of attribute values. Suppose we have created a class *Instructor*, and we have a Parquet file whose attributes match those of the class. Then we can read data from Parquet files as follows:

```
Dataset<Instructor> instructor = spark.read().parquet("...").
    as(Encoders.bean(Instructor.class));
```

In this case Parquet provides the names of attributes in the input file, which are used to map their values to attributes of the Instructor class. Unlike with Row, where the types are not known at compile time the types of attributes of Instructor are known at compile time, and can be represented more compactly than if we used the Row type. Further, the methods of the class Instructor can be used to access attributes; for example, we could use getSalary() instead of using col("salary"), which avoids the runtime cost of mapping attribute names to locations in the underlying records. More information on how to use these constructs can be found on the Spark documentation available online at spark.apache.org.

Our coverage of Spark has focused on database operations, but as mentioned earlier, Spark supports a number of other algebraic operations such as those related to machine learning, which can be invoked on DataSet types.

10.5 Streaming Data

Querying of data can be done in an ad hoc manner—for example, whenever an analyst wants to extract some information from a database. It can also be done in a periodic manner—for example, queries may be executed at the beginning of each day to get a summary of transactions that happened on the previous day.

However, there are many applications where queries need to be executed continuously, on data that arrive in a continuous fashion. The term **streaming data** refers to data that arrive in a continuous fashion. Many application domains need to process incoming data in real time (i.e., as they arrive, with delays, if any, guaranteed to be less than some bound).

10.5.1 Applications of Streaming Data

Here are a few examples of streaming data, and the real-time needs of applications that use the data.

- **Stock market**: In a stock market, each trade that is made (i.e., a stock is sold by someone and bought by someone else) is represented by a tuple. Trades are sent as a stream to processing systems.

  Stock market traders analyze the stream of trades to look for patterns, and they make buy or sell decisions based on the patterns that they observe. Real-time requirements in such systems used to be of the order of seconds in earlier days, but many of today's systems require delays to be of the order of tens of microseconds (usually to be able to react before others do, to the same patterns).

  Stock market regulators may use the same stream, but for a different purpose: to see if there are patterns of trades that are indicative of illegal activities. In both cases, there is a need for continuous queries that look for patterns; the results of the queries are used to carry out further actions.

- **E-commerce**: In an e-commerce site, each purchase made is represented as a tuple, and the sequence of all purchases forms a stream. Further, even the searches executed by a customer are of value to the e-commerce site, even if no actual purchase is made; thus, the searches executed by users form a stream.

  These streams can be used for multiple purposes. For example, if an advertising campaign is launched, the e-commerce site may wish to monitor in real time how the campaign is affecting searches and purchases. The e-commerce site may also wish to detect any spikes in sales of specific products to which it may need to respond by ordering more of that product. Similarly, the site may wish to monitor users for patterns of activities such as frequently returning items, and block further returns or purchases by the user.

- **Sensors**: Sensors are very widely used in systems such as vehicles, buildings, and factories. These sensors send readings periodically, and thus the readings form a stream. Readings in the stream are used to monitor the health of the system. If some readings are abnormal, actions may need to be taken to raise alarms, and to detect and fix any underlying faults, with minimal delay.

  Depending on the complexity of the system and the required frequency of readings, the stream can be of very high volume. In many cases, the monitoring is done at a central facility in the cloud, which monitors a very large number of systems. Parallel processing is essential in such a system to handle very large volumes of data in the incoming streams.

- **Network data**: Any organization that manages a large computer network needs to monitor activity on the network to detect network problems as well as attacks on the network by malicious software (malware). The underlying data being monitored can be represented as a stream of tuples containing data observed by each monitoring point (such as network switch). The tuples could contain information about individual network packets, such as source and destination addresses, size of the packet, and timestamp of packet generation. However, the rate of creation of tuples in such a stream is extremely high, and they cannot be handled except us-

ing special-purpose hardware. Instead, data can be aggregated to reduce the rate at which tuples are generated: for example, individual tuples could record data such as source and destination addresses, time interval, and total bytes transmitted in the time interval.

This aggregated stream must then be processed to detect problems. For example, link failures could be detected by observing a sudden drop in tuples traversing a particular link. Excessive traffic from multiple hosts to a single or a few destinations could indicate a denial-of-service attack. Packets sent from one host to many other hosts in the network could indicate malware trying to propagate itself to other hosts in the network. Detection of such patterns must be done in real time so that links can be fixed or action taken to stop the malware attack.

- **Social media**: Social media such as Facebook and Twitter get a continuous stream of messages (such as posts or tweets) from users. Each message in the stream of messages must be routed appropriately, for example by sending it to friends or followers. The messages that can potentially be delivered to a subscriber are then ranked and delivered in rank order, based on user preferences, past interactions, or advertisement charges.

  Social-media streams can be consumed not just by humans, but also by software. For example, companies may keep a lookout for tweets regarding the company and raise an alert if there are many tweets that reflect a negative sentiment about the company or its products. If a company launches an advertisement campaign, it may analyze tweets to see if the campaign had an impact on users.

There are many more examples of the need to process and query streaming data across a variety of domains.

10.5.2 Querying Streaming Data

Data stored in a database are sometimes referred to as **data-at-rest**, in contrast to streaming data. In contrast to stored data, streams are unbounded, that is, conceptually they may never end. Queries that can output results only after seeing all the tuples in a stream would then never be able to output any result. As an example, a query that asks for the number of tuples in a stream can never give a final result.

One way to deal with the unbounded nature of streams is to define **windows** on the streams, where each window contains tuples with a certain timestamp range or a certain number of tuples. Given information about timestamps of incoming tuples (e.g., they are increasing), we can infer when all tuples in a particular window have been seen. Based on the above, some query languages for streaming data require that windows be defined on streams, and queries can refer to one or a few windows of tuples rather than to a stream.

Another option is to output results that are correct at a particular point in the stream, but to output updates as more tuples arrive. For example, a count query can

output the number of tuples seen at a particular point in time, and as more tuples arrive, the query updates its result based on the new count.

Several approaches have been developed for querying streaming data, based on the two options described above.

1. **Continuous queries**. In this approach the incoming data stream is treated as inserts to a relation, and queries on the relations can be written in SQL, or using relational algebra operations. These queries can be registered as **continuous queries**, that is, queries that are running continuously. The result of the query on initial data are output when the system starts up. Each incoming tuple may result in insertion, update, or deletion of tuples in the result of the continuous query. The output of a continuous query is then a stream of updates to the query result, as the underlying database is updated by the incoming streams.

   This approach has some benefits in applications where users wish to view all database inserts that satisfy some conditions. However, a major drawback of the approach is that consumers of a query result would be flooded with a large number of updates to continuous queries if the input rate is high. In particular, this approach is not desirable for applications that output aggregated values, where users may wish to see final aggregates for some period of time, rather than every intermediate result as each incoming tuple is inserted.

2. **Stream query languages**. A second approach is to define a query language by extending SQL or relational algebra, where streams are treated differently from stored relations.

   Most stream query languages use window operations, which are applied to streams, and create relations corresponding to the contents of a window. For example, a window operation on a stream may create sets of tuples for each hour of the day; each such set is thus a relation. Relational operations can be executed on each such set of tuples, including aggregation, selection, and joins with stored relational data or with windows of other streams, to generate outputs.

   We provide an outline of stream query languages in Section 10.5.2.1. These languages separate streaming data from stored relations at the language level and require window operations to be applied before performing relational operations. Doing so ensures that results can be output after seeing only part of a stream. For example, if a stream guarantees that tuples have increasing timestamps, a window based on time can be deduced to have no more tuples once a tuple with a higher timestamp than the window end has been seen. The aggregation result for the window can be output at this point.

   Some streams do not guarantee that tuples have increasing timestamps. However, such streams would contain **punctuations**, that is, metadata tuples that state that all future tuples will have a timestamp greater than some value. Such punctuations are emitted periodically and can be used by window operators to decide

when an aggregate result, such as aggregates for an hourly window, is complete and can be output.

3. **Algebraic operators on streams**. A third approach is to allow users to write operators (user-defined functions) that are executed on each incoming tuple. Tuples are routed from inputs to operators; outputs of an operator may be routed to another operator, to a system output, or may be stored in a database. Operators can maintain internal state across the tuples that are processed, allowing them to aggregate incoming data. They may also be permitted to store data persistently in a database, for long-term use.

   This approach has seen widespread adoption in recent years, and we describe it in more detail later.

4. **Pattern matching**. A fourth option is to define a pattern matching language and allow users to write multiple rules, each with a pattern and an action. When the system finds a subsequence of tuples that match a particular pattern, the action corresponding to the pattern is executed. Such systems are called **complex event processing** (**CEP**) systems. Popular complex event processing systems include Oracle Event Processing, Microsoft StreamInsight, and FlinkCEP, which is part of the Apache Flink project,

We discuss stream query languages and algebraic operations in more detail later in this section.

Many stream-processing systems keep data in-memory and do not provide persistence guarantees; their goal is to generate results with minimum delay, to enable fast response based on analysis of streaming data. On the other hand, the incoming data may also need to be stored in a database for later processing. To support both patterns of querying, many applications use a so-called **lambda architecture**, where a copy of the input data is provided to the stream-processing system and another copy is provided to a database for storage and later processing. Such an architecture allows stream-processing systems to be developed rapidly, without worrying about persistence-related issues. However, the streaming system and database system are separate, resulting in these problems:

- Queries may need to be written twice, once for the streaming system and once for the database system, in different languages.
- Streaming queries may not be able to access stored data efficiently.

Systems that support streaming queries along with persistent storage and queries that span streams and stored data avoid these problems.

10.5.2.1 Stream Extensions to SQL

SQL window operations were described in Section 5.5.2, but stream query languages support further window types that are not supported by SQL window functions. For

example, a window that contains tuples for each hour cannot be specified using SQL window functions; note, however, that aggregates on such windows can be specified in SQL in a more roundabout fashion, first computing an extra attribute that contains just the hour component of a timestamp, and then grouping on the hour attribute. Window functions in streaming query languages simplify specification of such aggregation. Commonly supported window functions include:

- **Tumbling window**: Hourly windows are an example of tumbling windows. Windows do not overlap but are adjacent to each other. Windows are specified by their window size (for example, number of hours, minutes, or seconds).
- **Hopping window**: An hourly window computed every 20 minutes would be an example of a hopping window; the window width is fixed, similar to tumbling windows, but adjacent windows can overlap.
- **Sliding window**: Sliding windows are windows of a specified size (based on time, or number of tuples) around each incoming tuple. These are supported by the SQL standard.
- **Session window**: Session windows model users who perform multiple operations as part of a session. A window is identified by a user and a time-out interval, and contains a sequence of operations such that each operation occurs within the time-out interval from the previous operation. For example, if the time-out is 5 minutes, and a user performs an operation at 10 AM, a second operation at 10:04 AM, and a third operation at 11 AM, then the first two operations are part of one session, while the third is part of a different session. A maximum duration may also be specified, so once that duration expires, the session window is closed even if some operations have been performed within the time-out interval.

The exact syntax for specifying windows varies by implementation. Suppose we have a relation *order*(*orderid, datetime, itemid, amount*). In Azure Stream Analytics, the total order amount for each item for each hour can be specified by the following tumbling window:

```
select item, System.Timestamp as window_end, sum(amount)
from order timestamp by datetime
group by itemid, tumblingwindow(hour, 1)
```

Each output tuple has a timestamp whose value is equal to the timestamp of the end of the window; the timestamp can be accessed using the keyword *System.Timestamp* as shown in the query above.

SQL extensions to support streams differentiate between streams, where tuples have implicit timestamps and are expected to receive a potentially unbounded number of tuples and relations whose content is fixed at any point. For example, customers, suppliers, and items associated with orders would be treated as relations, rather than

as streams. The results of queries with windowing are treated as relations, rather than as streams.

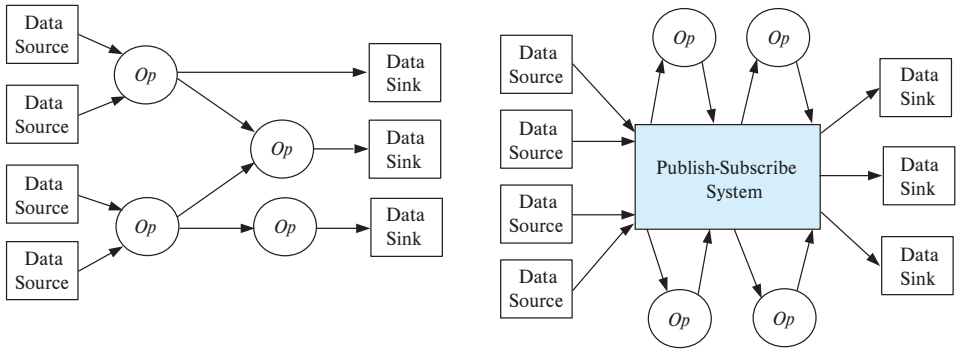
Joins are permitted between a stream and a relation, and the result is a stream; the timestamp of a join result tuple is the same as the timestamp of the input stream tuple. Joins between two streams have the problem that a tuple early in one stream may match a tuple that occurs much later in the other stream; such a join condition would require storing the entire stream for a potentially unbounded amount of time. To avoid this problem, streaming SQL systems allow stream-to-stream join only if there is a join condition that bounds the time difference between matching tuples. A condition that the timestamps of the two tuples differ by at most 1 hour is an example of such a condition.

10.5.3 Algebraic Operations on Streams

While SQL queries on streaming data are quite useful, there are many applications where SQL queries are not a good fit. With the algebraic operations approach to stream processing, user-defined code can be provided for implementing an algebraic operation; a number of predefined algebraic operations, such as selection and windowed aggregation, are also provided.

To perform computation, incoming tuples must be routed to operators that consume the tuples, and outputs of operators must be routed to their consumers. A key task of the implementation is to provide fault-tolerant routing of tuples between system input, operators, and outputs. *Apache Storm* and *Kafka* are widely used implementations that support such routing of data.

The *logical routing* of tuples is done by creating a directed acyclic graph (DAG) with operators as nodes. Edges between nodes define the flow of tuples. Each tuple output by an operator is sent along all the out-edges of the operator, to the consuming operators. Each operator receives tuples from all its in-edges. Figure 10.11a depicts the logical



(a) DAG representation of streaming data flow

(b) Publish-subscribe representation of streaming data flow

Figure 10.11 Routing of streams using DAG and publish-subscribe representations.

routing of stream tuples through a DAG structure. Operation nodes are denoted as "Op" nodes in the figure. The entry points to the stream-processing system are the data-source nodes of the DAG; these nodes consume tuples from the stream sources and inject them into the stream-processing system. The exit points of the stream-processing system are data-sink nodes; tuples exiting the system through a data sink may be stored in a data store or file system or may be output in some other manner.

One way of implementing a stream-processing system is by specifying the graph as part of the system configuration, which is read when the system starts processing tuples, and is then used to route tuples. The Apache Storm stream-processing system is an example of a system that uses a configuration file to define the graph, which is called a *topology* in the Storm system. Data-source nodes are called *spouts* in the Storm system, while operator nodes are called *bolts*, and edges connect these nodes.

An alternative way of creating such a routing graph is by using **publish-subscribe** systems. A publish-subscribe system allows publication of documents or other forms of data, with an associated topic. Subscribers correspondingly subscribe to specified topics. Whenever a document is published to a particular topic, a copy of the document is sent to all subscribers who have subscribed to that topic. Publish-subscribe systems are also referred to as **pub-sub** systems for short.

When a publish-subscribe system is used for routing tuples in a stream-processing system, tuples are considered documents, and each tuple is tagged with a topic. The entry points to the system conceptually "publish" tuples, each with an associated topic. Operators subscribe to one or more topics; the system routes all tuples with a specific topic to all subscribers of that topic. Operators can also publish their outputs back to the publish-subscribe system, with an associated topic.

A major benefit of the publish-subscribe approach is that operators can be added to the system, or removed from it, with relative ease. Figure 10.11b depicts the routing of tuples using a publish-subscribe representation. Each data source is assigned a unique topic name; the output of each operator is also assigned a unique topic name. Each operator subscribes to the topics of its inputs and publishes to the topics corresponding to its output. Data sources publish to their associated topic, while data sinks subscribe to the topics of the operators whose output goes to the sink.

The Apache Kafka system uses the publish-subscribe model to manage routing of tuples in streams. In the Kafka system, tuples published for a topic are retained for a specified period of time (called the retention period), even if there is currently no subscriber for the topic. Subscribers usually process tuples at the earliest possible time, but in case processing is delayed or temporarily stopped due to failures, the tuples are still available for processing until the retention time expires.

More details of routing, and in particular how publish-subscribe is implemented in a parallel system, are provided in Section 22.8.

The next detail to be addressed is how to implement the algebraic operations. We saw earlier how algebraic operations can be computed using data from files and other data sources as inputs.

Apache Spark allows streaming data sources to be used as inputs for such operations. The key issue is that some of the operations may not output any results at all until the entire stream is consumed, which may take potentially unbounded time. To avoid this problem, Spark breaks up streams into **discretized streams**, where the stream data for a particular time window are treated as a data input to algebraic operators. When the data in that window have been consumed, the operator generates its output, just as it would have if the data source were a file or a relation.

However, the above approach has the problem that the discretization of streams has to be done before any algebraic operations are executed. Other systems such as Apache Storm and Apache Flink support stream operations, which take a stream as input and output another stream. This is straightforward for operations such as map or relational select operations; each output tuple inherits a timestamp from the input tuple. On the other hand, relational aggregate operations and reduce operations may be unable to generate any output until the entire stream is consumed. To support such operations, Flink supports a window operation which breaks up the stream into windows; aggregates are computed within each window and are output once the window is complete. Note that the output is treated as a stream, where tuples have a timestamp based on the end of the window.[3]

10.6 Graph Databases

Graphs are an important type of data that databases need to deal with. For example, a computer network with multiple routers and links between them can be modeled as a graph, with routers as nodes and network links as edges. Road networks are another common type of graph, with road intersections modeled as nodes and the road links between intersections as edges. Web pages with hyperlinks between them are yet another example of graphs, where web pages can be modeled as nodes and hyperlinks between them as edges.

In fact, if we consider an E-R model of an enterprise, every entity can be modeled as a node of a graph, and every binary relationship can be modeled as an edge of the graph. Ternary and higher-degree relationships are harder to model, but as we saw in Section 6.9.4, such relationships can be modeled as a set of binary relationships if desired.

Graphs can be represented using the relational model using the following two relations:

1. node(ID, label, node_data)
2. edge(fromID, toID, label, edge_data)

where node_data and edge_data contain all the data related to nodes and edges, respectively.

[3]Some systems generate timestamps based on when the window is processed, but doing so results in output timestamps that are nondeterministic.

Modeling a graph using just two relations is too simplistic for complex database schemas. For example, applications require modeling of many types of nodes, each with its own set of attributes, and many types of edges, each with its own set of attributes. We can correspondingly have multiple relations that store nodes of different types and multiple relations that store edges of different types.

Although graph data can be easily stored in relational databases, graph databases such as the widely used Neo4j provide several extra features:

- They allow relations to be identified as representing nodes or edges and offer special syntax for defining such relations
- They support query languages designed for easily expressing path queries, which may be harder to express in SQL.
- They provide efficient implementations for such queries, which can execute queries much faster than if they were expressed in SQL and executed on a regular database.
- They provide support for other features such as graph visualization.

As an example of a graph query, we consider a query in the Cypher query language supported by Neo4j. Suppose the input graph has nodes corresponding to students (stored in a relation *student*) and instructors (stored in a relation *instructor*, and an edge type *advisor* from *student* to *instructor*. We omit details of how to create such node and edge types in Neo4j and assume appropriate schemas for these types. We can then write the following query:

```
match (i:instructor)<−[:advisor]−(s:student)
where i.dept_name= 'Comp. Sci.'
return i.ID as ID, i.name as name, collect(s.name) as advisees
```

Observe that the **match** clause in the query connects instructors to students via the advisor relation, which is modeled as a graph path that traverses the advisor edge in the backwards direction (the edge points from student to instructor), by using the syntax (*i*:*instructor*)<−[:*advisor*]−(*s*:*student*). This step basically performs a join of the instructor, advisor and student relations. The query then performs a group by on instructor ID and name, and collects all the students advised by the instructor into a set called advisees. We omit details, and refer the interested reader to online tutorials available at neo4j.com/developer.

Neo4J also supports recursive traversal of edges. For example, suppose we wish to find direct and indirect prerequisites of courses, with the relation *course* modeled with type node, and relation *prereq*(*course_id, prereq_id*) modeled with type edge. We can then write the following query:

```
match (c1:course)−[:prereq *1..]−>(c2:course)
return c1.course_id, c2.course_id
```

Here, the annotation "*1.." indicates we want to consider paths with multiple *prereq* edges, with a minimum of 1 edge (with a minimum of 0, a course would appear as its own prerequisite).

We note that Neo4j is a centralized system and does not (as of 2018) support parallel processing. However, there are many applications that need to process very large graphs, and parallel processing is key for such applications.

Computation of PageRank (which we saw earlier in Section 8.3.2.2) on very large graphs containing a node for every web page, and an edge for every hyperlink from one page to another, is a good example of a complex computation on very large graphs. The web graph today has hundreds of billions of nodes and trillions of edges. Social networks are another example of very large graphs, containing billions of nodes and edges; computations on such graphs include shortest paths (to find connectivity between people), or computing how influential people are based on edges in the social network.

There are two popular approaches for parallel graph processing:

1. **Map-reduce and algebraic frameworks**: Graphs can be represented as relations, and individual steps of many parallel graph algorithms can be represented as joins. Graphs can thus be stored in a parallel storage system, partitioned across multiple machines. We can then use map-reduce programs, algebraic frameworks such as Spark, or parallel relational database implementations to process each step of a graph algorithm in parallel across multiple nodes.

   Such approaches work well for many applications. However, when performing iterative computations that traverse long paths in graphs, these approaches are quite inefficient, since they typically read the entire graph in each iteration.

2. **Bulk synchronous processing frameworks**: The **bulk synchronous processing** (**BSP**) framework for graph algorithms frames graph algorithms as computations associated with vertices that operate in an iterative manner. Unlike the preceding approach, here the graph is typically stored in memory, with vertices partitioned across multiple machines; most importantly, the graph does not have to be read in each iteration.

   Each vertex (node) of the graph has data (state) associated with it. Similar to how programmers provide `map()` and `reduce()` functions in the MapReduce framework, in the BSP framework programmers provide methods that are executed for each node of the graph. The methods can send messages to neighboring nodes and receive messages from neighboring nodes of the graph. In each iteration, called a **superstep**, the method associated with each node is executed; the method consumes any incoming messages, updates the data associated with the node, and may optionally send messages to neighboring nodes. Messages sent in one iteration are received by the recipients in the next iteration. The method executing at each vertex may vote to halt if they decide they have no more computation to carry out. If in some iteration all vertices vote to halt, and no messages are sent out, the computation can be halted.

The result of the computation is contained in the state at each node. The state can be collected and output as the result of the computation.

The idea of bulk synchronous processing is quite old but was popularized by the *Pregel* system developed by Google, which provided a fault-tolerant implementation of the framework. The *Apache Giraph* system is an open-source version of the Pregel system.

The GraphX component of Apache Spark supports graph computations on large graphs. It provides an API based on Pregel, as well as a number of other operations that take a graph as input, and output a graph. Operations supported by GraphX include map functions applied on vertices and edges of graphs, join of a graph with an RDD, and an aggregation operation that works as follows: a user-defined function is used to create messages that are sent to all the neighbors of each node, and another user-defined function is used to aggregate the messages. All these operations can be executed in parallel to handle large graphs.

For more information on how to write graph algorithms in such settings, see the references in the Further Reading section at the end of the chapter.

10.7 Summary

- Modern data management applications often need to deal with data that are not necessarily in relational form, and these applications also need to deal with volumes of data that are far larger than what a single traditional organization would have generated.
- The increasing use of data sensors leads to the connection of sensors and other computing devices embedded within other objects to the internet, often referred to as the "internet of things."
- There is now a wider variety of query language options for Big Data applications, driven by the need to handle more varied of data types, and by the need to scale to very large data volumes/velocity.
- Building data management systems that can scale to large volume/velocity of data requires parallel storage and processing of data.
- Distributed file systems allow files to be stored across a number of machines, while allowing access to files using a traditional file-system interface.
- Key-value storage systems allow records to be stored and retrieved based on a key and may additionally provide limited query facilities. These systems are not full-fledged database systems, and they are sometimes called NoSQL systems.
- Parallel and distributed databases provide a traditional database interface, but they store data across multiple machines, and they perform query processing in parallel across multiple machines.

- The MapReduce paradigm models a common situation in parallel processing, where some processing, identified by the `map()` function, is applied to each of a large number of input records, and then some form of aggregation, identified by the `reduce()` function, is applied to the result of the `map()` function.
- The Hadoop system provides a widely used open-source implementation of MapReduce in the Java language.
- There are a large number of applications that use the MapReduce paradigm for data processing of various kinds whose logic can be easily expressed using SQL.
- Relational algebra forms the foundation of relational query processing, allowing queries to be modeled as trees of operations. This idea is extended to settings with more complex data types by supporting algebraic operators that can work on datasets containing records with complex data types, and returning datasets with records containing similar complex data types.
- There are many applications where queries need to be executed continuously, on data that arrive in a continuous fashion. The term **streaming data** refers to data that arrive in a continuous fashion. Many application domains need to process incoming data in real time.
- Graphs are an important type of data that databases need to deal with.

Review Terms

- Volume
- Velocity
- Conversion
- Internet of things
- Distributed file system
- NameNode server
- DataNodes machines
- Sharding
- Partitioning attribute
- Key-value storage system
- Key-value store
- Document stores
- NoSQL systems
- Shard key
- Parallel databases
- Reduce key
- Shuffle step
- Streaming data
- Data-at-rest
- Windows on the streams
- Continuous queries
- Punctuations
- Lambda architecture
- Tumbling window
- Hopping window
- Sliding window
- Session window
- Publish-subscribe systems
- Pub-sub systems
- Discretized streams
- Superstep

Practice Exercises

10.1 Suppose you need to store a very large number of small files, each of size say 2 kilobytes. If your choice is between a distributed file system and a distributed key-value store, which would you prefer, and explain why.

10.2 Suppose you need to store data for a very large number of students in a distributed document store such as MongoDB. Suppose also that the data for each student correspond to the data in the *student* and the *takes* relations. How would you represent the above data about students, ensuring that all the data for a particular student can be accessed efficiently? Give an example of the data representation for one student.

10.3 Suppose you wish to store utility bills for a large number of users, where each bill is identified by a customer ID and a date. How would you store the bills in a key-value store that supports range queries, if queries request the bills of a specified customer for a specified date range.

10.4 Give pseudocode for computing a join $r \bowtie_{r.A=s.A} s$ using a single MapReduce step, assuming that the `map()` function is invoked on each tuple of r and s. Assume that the `map()` function can find the name of the relation using `context.relname()`.

10.5 What is the conceptual problem with the following snippet of Apache Spark code meant to work on very large data. Note that the `collect()` function returns a Java collection, and Java collections (from Java 8 onwards) support map and reduce functions.

```
JavaRDD<String< lines = sc.textFile("logDirectory");
int totalLength = lines.collect().map(s -> s.length())
                    .reduce(0,(a,b) -> a+b);
```

10.6 Apache Spark:

a. How does Apache Spark perform computations in parallel?

b. Explain the statement: "Apache Spark performs transformations on RDDs in a lazy manner."

c. What are some of the benefits of lazy evaluation of operations in Apache Spark?

10.7 Given a collection of documents, for each word w_i, let n_i denote the number of times the word occurs in the collection. Let N be the total number of word occurrences across all documents. Next, consider all pairs of consecutive words

(w_i, w_j) in the document; let $n_{i,j}$ denote the number of occurrences of the word pair (w_i, w_j) across all documents.

Write an Apache Spark program that, given a collection of documents in a directory, computes N, all pairs (w_i, n_i), and all pairs $((w_i, w_j), n_{i,j})$. Then output all word pairs such that $n_{i,j}/N \geq 10 * (n_i/N) * (n_j/N)$. These are word pairs that occur 10 times or more as frequently as they would be expected to occur if the two words occurred independently of each other.

You will find the join operation on RDDs useful for the last step, to bring related counts together. For simplicity, do not bother about word pairs that cross lines. Also assume for simplicity that words only occur in lowercase and that there are no punctuation marks.

10.8 Consider the following query using the tumbling window operator:

select *item*, *System.Timestamp* **as** *window_end*, **sum**(*amount*)
from *order* **timestamp by** *datetime*
group by *itemid*, **tumblingwindow**(*hour*, 1)

Give an equivalent query using normal SQL constructs, without using the tumbling window operator. You can assume that the timestamp can be converted to an integer value that represents the number of seconds elapsed since (say) midnight, January 1, 1970, using the function *to_seconds*(*timestamp*). You can also assume that the usual arithmetic functions are available, along with the function *floor*(a) which returns the largest integer $\leq a$.

10.9 Suppose you wish to model the university schema as a graph. For each of the following relations, explain whether the relation would be modeled as a node or as an edge:

(i) *student*, (ii) *instructor*, (iii) *course*, (iv) *section*, (v) *takes*, (vi) *teaches*.

Does the model capture connections between sections and courses?

Exercises

10.10 Give four ways in which information in web logs pertaining to the web pages visited by a user can be used by the web site.

10.11 One of the characteristics of Big Data is the variety of data. Explain why this characteristic has resulted in the need for languages other than SQL for processing Big Data.

10.12 Suppose your company has built a database application that runs on a centralized database, but even with a high-end computer and appropriate indices created on the data, the system is not able to handle the transaction load, lead-

ing to slow processing of queries. What would be some of your options to allow the application to handle the transaction load?

10.13 The map-reduce framework is quite useful for creating inverted indices on a set of documents. An inverted index stores for each word a list of all document IDs that it appears in (offsets in the documents are also normally stored, but we shall ignore them in this question).

For example, if the input document IDs and contents are as follows:

1: data clean
2: data base
3: clean base

then the inverted lists would

data: 1, 2
clean: 1, 3
base: 2, 3

Give pseudocode for map and reduce functions to create inverted indices on a given set of files (each file is a document). Assume the document ID is available using a function context.getDocumentID(), and the map function is invoked once per line of the document. The output inverted list for each word should be a list of document IDs separated by commas. The document IDs are normally sorted, but for the purpose of this question you do not need to bother to sort them.

10.14 Fill in the blanks below to complete the following Apache Spark program which computes the number of occurrences of each word in a file. For simplicity we assume that words only occur in lowercase, and there are no punctuation marks.

```
JavaRDD<String> textFile = sc.textFile("hdfs://...");
JavaPairRDD<String, Integer> counts =
    textFile._____(s -> Arrays.asList(s.split(" ")).______())
     .mapToPair(word -> new _____).reduceByKey((a, b) -> a + b);
```

10.15 Suppose a stream can deliver tuples out of order with respect to tuple timestamps. What extra information should the stream provide, so a stream query processing system can decide when all tuples in a window have been seen?

10.16 Explain how multiple operations can be executed on a stream using a publish-subscribe system such as Apache Kafka.

Tools

A wide variety of open-source Big Data tools are available, in addition to some commercial tools. In addition, a number of these tools are available on cloud plat-

forms. We list below several popular tools, along with the URLs where they can be found. Apache HDFS (hadoop.apache.org) is a widely used distributed file system implementation. Open-source distributed/parallel key-value stores include Apache HBase (hbase.apache.org), Apache Cassandra (cassandra.apache.org), MongoDB (www.mongodb.com), and Riak (basho.com).

Hosted cloud storage systems include the Amazon S3 storage system (aws.amazon.com/s3) and Google Cloud Storage (cloud.google.com/storage). Hosted key-value stores include Google BigTable (cloud.google.com/bigtable), and Amazon DynamoDB (aws.amazon.com/dynamodb).

Google Spanner (cloud.google.com/spanner) and the open source CockroachDB (www.cockroachlabs.com) are scalable parallel databases that support SQL and transactions, and strongly consistent storage.

Open-source MapReduce systems include Apache Hadoop (hadoop.apache.org), and Apache Spark (spark.apache.org), while Apache Tez (tez.apache.org) supports data processing using a DAG of algebraic operators. These are also available as cloud-based offerings from Amazon Elastic MapReduce (aws.amazon.com/emr), which also supports Apache HDFS and Apache HBase, and from Microsoft Azure (azure.microsoft.com).

Apache Hive (hive.apache.org) is a popular open-source SQL implementation that runs on top of the Apache MapReduce, Apache Tez, as well as Apache Spark; these systems are designed to support large queries running in parallel on multiple machines. Apache Impala (impala.apache.org) is an SQL implementation that runs on Hadoop, and is designed to handle a large number of queries, and to return query results with minimal delays (latency). Hosted SQL offerings on the cloud that support parallel processing include Amazon EMR (aws.amazon.com/emr), Google Cloud SQL (cloud.google.com/sql), and Microsoft Azure SQL (azure.microsoft.com).

Apache Kafka (kafka.apache.org) and Apache Flink (flink.apache.org) are open-source stream-processing systems; Apache Spark also provides support for stream processing. Hosted stream-processing platforms include Amazon Kinesis (aws.amazon.com/kinesis), Google Cloud Dataflow (cloud.google.com/dataflow) and Microsoft Stream Analytics (azure.microsoft.com). Open source graph processing platforms include Neo4J (neo4j.com) and Apache Giraph (giraph.apache.org).

Further Reading

[Davoudian et al. (2018)] provides a nice survey of NoSQL data stores, including data model querying and internals. More information about Apache Hadoop, including documentation on HDFS and Hadoop MapReduce, can be found on the Apache Hadoop homepage, hadoop.apache.org. Information about Apache Spark may be found on the Spark homepage, spark.apache.org. Information about the Apache Kafka streaming data platform may be found on kafka.apache.org, and details of stream processing in Apache Flink may be found on flink.apache.org. Bulk Synchronous Processing

was introduced in [Valiant (1990)]. A description of the Pregel system, including its support for bulk synchronous processing, may be found in [Malewicz et al. (2010)], while information about the open source equivalent, Apache Giraph, may be found on giraph.apache.org.

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Credits

The photo of the sailboats in the beginning of the chapter is due to ©Pavel Nesvadba/Shutterstock.

CHAPTER 11

Data Analytics

Decision-making tasks benefit greatly by using data about the past to predict the future and using the predictions to make decisions. For example, online advertisement systems need to decide what advertisement to show to each user. Analysis of past actions and profiles of other users, as well as past actions and profile of the current user, are key to deciding which advertisement the user is most likely to respond to. Here, each decision is low-value, but with high volumes the overall value of making the right decisions is very high. At the other end of the value spectrum, manufacturers and retailers need to decide what items to manufacture or stock many months ahead of the actual sale of the items. Predicting future demand of different types of items based on past sales and other indicators is key to avoiding both overproduction or overstocking of some items, and underproduction or understocking of other items. Errors can lead to monetary loss due to unsold inventory of some items, or loss of potential revenue due to nonavailability of some items.

The term **data analytics** refers broadly to the processing of data to infer patterns, correlations, or models for prediction. The results of analytics are then used to drive business decisions.

The financial benefits of making correct decisions can be substantial, as can the costs of making wrong decisions. Organizations therefore invest a lot of money to gather or purchase required data and build systems for data analytics.

11.1 Overview of Analytics

Large companies have diverse sources of data that they need to use for making business decisions. The sources may store the data under different schemas. For performance reasons (as well as for reasons of organization control), the data sources usually will not permit other parts of the company to retrieve data on demand.

Organizations therefore typically gather data from multiple sources into one location, referred to as a *data warehouse*. Data warehouses gather data from multiple sources at a single site, under a unified schema (which is usually designed to support efficient analysis, even at the cost of redundant storage). Thus, they provide the user

a single uniform interface to data. However, data warehouses today also collect and store data from non-relational sources, where schema unification is not possible. Some sources of data have errors that can be detected and corrected using business constraints; further, organizations may collect data from multiple sources, and there may be duplicates in the data collected from different sources. These steps of collecting data, cleaning/deduplicating the data, and loading the data into a warehouse are referred to as *extract, transform and load* (ETL) tasks. We study issues in building and maintaining a data warehouse in Section 11.2.

The most basic form of analytics is generation of aggregates and reports summarizing the data in ways that are meaningful to the organization. Analysts need to get a number of different aggregates and compare them to understand patterns in the data. Aggregates, and in some cases the underlying data, are typically presented graphically as charts, to make it easy for humans to visualize the data. Dashboards that display charts summarizing key organizational parameters, such as sales, expenses, product returns, and so forth, are popular means of monitoring the health of an organization. Analysts also need to visualize data in ways that can highlight anomalies or give insights into causes for changes in the business.

Systems that support very efficient analysis, where aggregate queries on large data are answered in almost real time (as opposed to being answered after tens of minutes or multiple hours) are popular with analysts. Such systems are referred to as *online analytical processing* (OLAP) systems. We discuss online analytical processing in Section 11.3, where we cover the concept of multidimensional data, OLAP operations, relational representation of multidimensional summaries. We also discuss graphical representation of data and visualization in Section 11.3.

Statistical analysis is an important part of data analysis. There are several tools that are designed for statistical analysis, including the R language/environment, which is open source, and commercial systems such as SAS and SPSS. The R language is widely used today, and in addition to features for statistical analysis, it supports facilities for graphical display of data. A large number of R packages (libraries) are available that implement a wide variety of data analysis tasks, including many machine-learning algorithms. R has been integrated with databases as well as with Big Data systems such as Apache Spark, which allows R programs to be executed in parallel on large datasets. Statistical analysis is a large area by itself, and we do not discuss it further in this book. References providing more information may be found in the Further Reading section at the end of this chapter.

Prediction of different forms is another key aspect of analytics. For example, banks need to decide whether to give a loan to a loan applicant, and online advertisers need to decide which advertisement to show to a particular user. As another example, manufacturers and retailers need to make decisions on what items to manufacture or order in what quantities.

These decisions are driven significantly by techniques for analyzing past data and using the past to predict the future. For example, the risk of loan default can be predicted as follows. First, the bank would examine the loan default history of past cus-

tomers, find key features of customers, such as salary, education level, job history, and so on, that help in prediction of loan default. The bank would then build a prediction model (such as a decision tree, which we study later in this chapter) using the chosen features. When a customer then applies for a loan, the features of that particular customer are fed into the model which makes a prediction, such as an estimated likelihood of loan default. The prediction is used to make business decisions, such as whether to give a loan to the customer.

Similarly, analysts may look at the past history of sales and use it to predict future sales, to make decisions on what and how much to manufacture or order, or how to target their advertising. For example, a car company may search for customer attributes that help predict who buys different types of cars. It may find that most of its small sports cars are bought by young women whose annual incomes are above $50,000. The company may then target its marketing to attract more such women to buy its small sports cars and may avoid wasting money trying to attract other categories of people to buy those cars.

Machine-learning techniques are key to finding patterns in data and in making predictions from these patterns. The field of *data mining* combines knowledge-discovery techniques invented by machine-learning researchers with efficient implementation techniques that enable them to be used on extremely large databases. Section 11.4 discusses data mining.

The term **business intelligence** (**BI**) is used in a broadly similar sense to data analytics. The term **decision support** is also used in a related but narrower sense, with a focus on reporting and aggregation, but not including machine learning/data mining. Decision-support tasks typically use SQL queries to process large amounts of data. Decision-support queries are can be contrasted with queries for *online transaction processing*, where each query typically reads only a small amount of data and may perform a few small updates.

11.2 Data Warehousing

Large organizations have a complex internal organization structure, and therefore different data may be present in different locations, or on different operational systems, or under different schemas. For instance, manufacturing-problem data and customer-complaint data may be stored on different database systems. Organizations often purchase data from external sources, such as mailing lists that are used for product promotions, or credit scores of customers that are provided by credit bureaus, to decide on creditworthiness of customers.[1]

Corporate decision makers require access to information from multiple such sources. Setting up queries on individual sources is both cumbersome and inefficient.

[1] Credit bureaus are companies that gather information about consumers from multiple sources and compute a creditworthiness score for each consumer.

Moreover, the sources of data may store only current data, whereas decision makers may need access to past data as well; for instance, information about how purchase patterns have changed in the past few years could be of great importance. Data warehouses provide a solution to these problems.

A **data warehouse** is a repository (or archive) of information gathered from multiple sources, stored under a unified schema, at a single site. Once gathered, the data are stored for a long time, permitting access to historical data. Thus, data warehouses provide the user a single consolidated interface to data, making decision-support queries easier to write. Moreover, by accessing information for decision support from a data warehouse, the decision maker ensures that online transaction-processing systems are not affected by the decision-support workload.

11.2.1 Components of a Data Warehouse

Figure 11.1 shows the architecture of a typical data warehouse and illustrates the gathering of data, the storage of data, and the querying and data analysis support. Among the issues to be addressed in building a warehouse are the following:

- **When and how to gather data.** In a **source-driven architecture** for gathering data, the data sources transmit new information, either continually (as transaction processing takes place), or periodically (nightly, for example). In a **destination-driven architecture**, the data warehouse periodically sends requests for new data to the sources.

  Unless updates at the sources are "synchronously" replicated at the warehouse, the warehouse will never be quite up-to-date with the sources. Synchronous replication can be expensive, so many data warehouses do not use synchronous replication, and they perform queries only on data that are old enough that they have

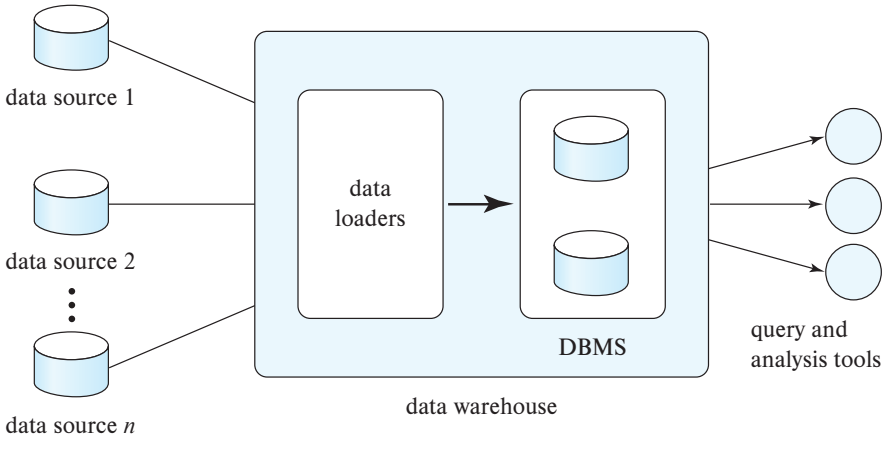


Figure 11.1 Data-warehouse architecture.

been completely replicated. Traditionally, analysts were happy with using yesterday's data, so data warehouses could be loaded with data up to the end of the previous day. However, increasingly organizations want more up-to-date data. The data freshness requirements depend on the application. Data that are within a few hours old may be sufficient for some applications; others that require real-time responses to events may use stream processing infrastructure (described in Section 10.5) instead of depending on a warehouse infrastructure.

- **What schema to use.** Data sources that have been constructed independently are likely to have different schemas. In fact, they may even use different data models. Part of the task of a warehouse is to perform schema integration and to convert data to the integrated schema before they are stored. As a result, the data stored in the warehouse are not just a copy of the data at the sources. Instead, they can be thought of as a materialized view of the data at the sources.

- **Data transformation and cleansing.** The task of correcting and preprocessing data is called **data cleansing**. Data sources often deliver data with numerous minor inconsistencies, which can be corrected. For example, names are often misspelled, and addresses may have street, area, or city names misspelled, or postal codes entered incorrectly. These can be corrected to a reasonable extent by consulting a database of street names and postal codes in each city. The approximate matching of data required for this task is referred to as **fuzzy lookup**.

  Address lists collected from multiple sources may have duplicates that need to be eliminated in a **merge–purge operation** (this operation is also referred to as **deduplication**). Records for multiple individuals in a house may be grouped together so only one mailing is sent to each house; this operation is called **householding**.

  Data may be **transformed** in ways other than cleansing, such as changing the units of measure, or converting the data to a different schema by joining data from multiple source relations. Data warehouses typically have graphical tools to support data transformation. Such tools allow transformation to be specified as boxes, and edges can be created between boxes to indicate the flow of data. Conditional boxes can route data to an appropriate next step in transformation.

- **How to propagate updates**. Updates on relations at the data sources must be propagated to the data warehouse. If the relations at the data warehouse are exactly the same as those at the data source, the propagation is straightforward. If they are not, the problem of propagating updates is basically the *view-maintenance* problem, which was discussed in Section 4.2.3, and is covered in more detail in Section 16.5.

- **What data to summarize.** The raw data generated by a transaction-processing system may be too large to store online. However, we can answer many queries by maintaining just summary data obtained by aggregation on a relation, rather than maintaining the entire relation. For example, instead of storing data about every sale of clothing, we can store total sales of clothing by item name and category.

The different steps involved in getting data into a data warehouse are called **extract, transform, and load** or **ETL** tasks; extraction refers to getting data from the sources, while load refers to loading the data into the data warehouse. In current generation data warehouses that support user-defined functions or MapReduce frameworks, data may be extracted, loaded into the warehouse, and then transformed. The steps are then referred to as **extract, load, and transform** or **ELT** tasks. The ELT approach permits the use of parallel processing frameworks for data transformation.

11.2.2 Multidimensional Data and Warehouse Schemas

Data warehouses typically have schemas that are designed for data analysis, using tools such as OLAP tools. The relations in a data **warehouse schema** can usually be classified as *fact tables* and *dimension tables*. **Fact tables** record information about individual events, such as sales, and are usually very large. A table recording sales information for a retail store, with one tuple for each item that is sold, is a typical example of a fact table. The attributes in fact table can be classified as either *dimension attributes* or *measure attributes*, The **measure attributes** store quantitative information, which can be aggregated upon; the measure attributes of a *sales* table would include the number of items sold and the price of the items. In contrast, **dimension attributes** are dimensions upon which measure attributes, and summaries of measure attributes, are grouped and viewed. The dimension attributes of a *sales* table would include an item identifier, the date when the item is sold, which location (store) the item was sold from, the customer who bought the item, and so on.

Data that can be modeled using dimension attributes and measure attributes are called **multidimensional data**.

To minimize storage requirements, dimension attributes are usually short identifiers that are foreign keys into other tables called **dimension tables**. For instance, a fact table *sales* would have dimension attributes *item_id*, *store_id*, *customer_id*, and *date*, and measure attributes *number* and *price*. The attribute *store_id* is a foreign key into a dimension table *store*, which has other attributes such as store location (city, state, country). The *item_id* attribute of the *sales* table would be a foreign key into a dimension table *item_info*, which would contain information such as the name of the item, the category to which the item belongs, and other item details such as color and size. The *customer_id* attribute would be a foreign key into a *customer* table containing attributes such as name and address of the customer. We can also view the *date* attribute as a foreign key into a *date_info* table giving the month, quarter, and year of each date.

The resultant schema appears in Figure 11.2. Such a schema, with a fact table, multiple dimension tables, and foreign keys from the fact table to the dimension tables is called a **star schema**. More complex data-warehouse designs may have multiple levels of dimension tables; for instance, the *item_info* table may have an attribute *manufacturer_id* that is a foreign key into another table giving details of the manufacturer. Such schemas are called **snowflake schemas**. Complex data-warehouse designs may also have more than one fact table.

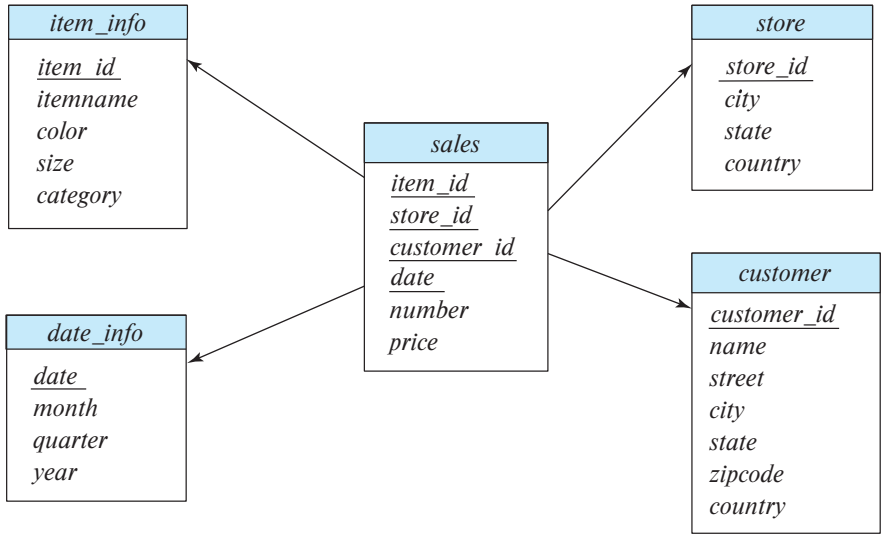


Figure 11.2 Star schema for a data warehouse.

11.2.3 Database Support for Data Warehouses

The requirements of a database system designed for transaction processing are somewhat different from one designed to support a data-warehouse system. One key difference is that a transaction-processing database needs to support many small queries, which may involve updates in addition to reads. In contrast, data warehouses typically need to process far fewer queries, but each query accesses a much larger amount of data.

Most importantly, while new records are inserted into relations in a data warehouse, and old records may be deleted once they are no longer needed, to make space for new data, records are typically never updated once they are added to a relation. Thus, data warehouses do not need to pay any overhead for concurrency control. (As described in Chapter 17 and Chapter 18, if concurrent transactions read and write the same data, the resultant data may become inconsistent. Concurrency control restricts concurrent accesses in a way that ensures there is no erroneous update to the database.) The overhead of concurrency control can be significant in terms of not just time taken for query processing, but also in terms of storage, since databases often store multiple versions of data to avoid conflicts between small update transactions and long read-only transactions. None of these overheads are needed in a data warehouse.

Databases traditionally store all attributes of a tuple together, and tuples are stored sequentially in a file. Such a storage layout is referred to as *row-oriented storage*. In contrast, in **column-oriented storage**, each attribute of a relation is stored in a separate file, with values from successive tuples stored at successive positions in the file. Assuming fixed-size data types, the value of attribute A of the ith tuple of a relation can be found

by accessing the file corresponding to attribute A and reading the value at offset $(i - 1)$ times the size (in bytes) of values in attribute A.

Column-oriented storage has at least two major benefits over row-oriented storage:

1. When a query needs to access only a few attributes of a relation with a large number of attributes, the remaining attributes need not be fetched from disk into memory. In contrast, in row-oriented storage, not only are irrelevant attributes fetched into memory, but they may also get prefetched into processor cache, wasting cache space and memory bandwidth, if they are stored adjacent to attributes used in the query.
2. Storing values of the same type together increases the effectiveness of compression; compression can greatly reduce both the disk storage cost and the time to retrieve data from disk.

On the other hand, column-oriented storage has the drawback that storing or fetching a single tuple requires multiple I/O operations.

As a result of these trade-offs, column-oriented storage is not widely used for transaction-processing applications. However, column-oriented storage is today widely used for data-warehousing applications, where accesses are rarely to individual tuples but rather require scanning and aggregating multiple tuples. Column-oriented storage is described in more detail in Section 13.6.

Database implementations that are designed purely for data warehouse applications include Teradata, Sybase IQ, and Amazon Redshift. Many traditional databases support efficient execution of data warehousing applications by adding features such as columnar storage; these include Oracle, SAP HANA, Microsoft SQL Server, and IBM DB2.

In the 2010s there has been an explosive growth in Big Data systems that are designed to process queries over data stored in files. Such systems are now a key part of the data warehouse infrastructure. As we saw in Section 10.3, the motivation for such systems was the growth of data generated by online systems in the form of log files, which have a lot of valuable information that can be exploited for decision support. However, these systems can handle any kind of data, including relational data. Apache Hadoop is one such system, and the Hive system allows SQL queries to be executed on top of the Hadoop system.

A number of companies provide software to optimize Hive query processing, including Cloudera and Hortonworks. Apache Spark is another popular Big Data system that supports SQL queries on data stored in files. Compressed file structures containing records with columns, such as Orc and Parquet, are increasingly used to store such log records, simplifying integration with SQL. Such file formats are discussed in more detail in Section 13.6.

11.2.4 Data Lakes

While data warehouses pay a lot of attention to ensuring a common data schema to ease the job of querying the data, there are situations where organizations want to store data without paying the cost of creating a common schema and transforming data to the common schema. The term **data lake** is used to refer to a repository where data can be stored in multiple formats, including structured records and unstructured file formats. Unlike data warehouses, data lakes do not require up-front effort to preprocess data, but they do require more effort when creating queries. Since data may be stored in many different formats, querying tools also need to be quite flexible. Apache Hadoop and Apache Spark are popular tools for querying such data, since they support querying of both unstructured and structured data.

11.3 Online Analytical Processing

Data analysis often involves looking for patterns that arise when data values are grouped in "interesting" ways. As a simple example, summing credit hours for each department is a way to discover which departments have high teaching responsibilities. In a retail business, we might group sales by product, the date or month of the sale, the color or size of the product, or the profile (such as age group and gender) of the customer who bought the product.

11.3.1 Aggregation on Multidimensional Data

Consider an application where a shop wants to find out what kinds of clothes are popular. Let us suppose that clothes are characterized by their item_name, color, and size, and that we have a relation *sales* with the schema.

$$sales\ (item\_name, color, clothes\_size, quantity)$$

Suppose that *item_name* can take on the values (skirt, dress, shirt, pants), *color* can take on the values (dark, pastel, white), *clothes_size* can take on values (small, medium, large), and *quantity* is an integer value representing the total number of items of a given {*item_name*, *color*, *clothes_size*}. An instance of the *sales* relation is shown in Figure 11.3.

Statistical analysis often requires grouping on multiple attributes. The attribute *quantity* of the *sales* relation is a measure attribute, since it measures the number of units sold, while *item_name*, *color*, and *clothes_size* are dimension attributes. (A more realistic version of the *sales* relation would have additional dimensions, such as time and sales location, and additional measures such as monetary value of the sale.)

To analyze the multidimensional data, a manager may want to see data laid out as shown in the table in Figure 11.4. The table shows total quantities for different combinations of *item_name* and *color*. The value of *clothes_size* is specified to be **all**, indicating that the displayed values are a summary across all values of *clothes_size* (i.e., we want to group the "small," "medium," and "large" items into one single group.

item_name	*color*	*clothes_size*	*quantity*
dress	dark	small	2
dress	dark	medium	6
dress	dark	large	12
dress	pastel	small	4
dress	pastel	medium	3
dress	pastel	large	3
dress	white	small	2
dress	white	medium	3
dress	white	large	0
pants	dark	small	14
pants	dark	medium	6
pants	dark	large	0
pants	pastel	small	1
pants	pastel	medium	0
pants	pastel	large	1
pants	white	small	3
pants	white	medium	0
pants	white	large	2
shirt	dark	small	2
shirt	dark	medium	6
shirt	dark	large	6
shirt	pastel	small	4
shirt	pastel	medium	1
shirt	pastel	large	2
shirt	white	small	17
shirt	white	medium	1
shirt	white	large	10
skirt	dark	small	2
skirt	dark	medium	5
skirt	dark	large	1
skirt	pastel	small	11
skirt	pastel	medium	9
skirt	pastel	large	15
skirt	white	small	2
skirt	white	medium	5
skirt	white	large	3

Figure 11.3 An example of *sales* relation.

The table in Figure 11.4 is an example of a **cross-tabulation** (or **cross-tab**, for short), also referred to as a **pivot-table**. In general, a cross-tab is a table derived from a relation

clothes_size **all**

item_name	*color*			
	dark	pastel	white	total
skirt	8	35	10	53
dress	20	10	5	35
shirt	14	7	28	49
pants	20	2	5	27
total	62	54	48	164

Figure 11.4 Cross-tabulation of *sales* by *item_name* and *color*.

(say R), where values for one attribute (say A) form the column headers and values for another attribute (say B) form the row header. For example, in Figure 11.4, the attribute *color* corresponds to A (with values "dark," "pastel," and "white"), and the attribute *item_name* corresponds to B (with values "skirt," "dress," "shirt," and "pants").

Each cell in the pivot-table can be identified by (a_i, b_j), where a_i is a value for A and b_j a value for B. The values of the various cells in the pivot-table are derived from the relation R as follows: If there is at most one tuple in R with any (a_i, b_j) value, the value in the cell is derived from that single tuple (if any); for instance, it could be the value of one or more other attributes of the tuple. If there can be multiple tuples with an (a_i, b_j) value, the value in the cell must be derived by aggregation on the tuples with that value. In our example, the aggregation used is the sum of the values for attribute *quantity*, across all values for *clothes_size*, as indicated by "*clothes_size*: **all**" above the cross-tab in Figure 11.4. Thus, the value for cell (skirt, pastel) is 35, since there are three tuples in the *sales* table that meet that criteria, with values 11, 9, and 15.

In our example, the cross-tab also has an extra column and an extra row storing the totals of the cells in the row/column. Most cross-tabs have such summary rows and columns.

The generalization of a cross-tab, which is two-dimensional, to n dimensions can be visualized as an n-dimensional cube, called the **data cube**. Figure 11.5 shows a data cube on the *sales* relation. The data cube has three dimensions, *item_name*, *color*, and *clothes_size*, and the measure attribute is *quantity*. Each cell is identified by values for these three dimensions. Each cell in the data cube contains a value, just as in a cross-tab. In Figure 11.5, the value contained in a cell is shown on one of the faces of the cell; other faces of the cell are shown blank if they are visible. All cells contain values, even if they are not visible. The value for a dimension may be **all**, in which case the cell contains a summary over all values of that dimension, as in the case of cross-tabs.

The number of different ways in which the tuples can be grouped for aggregation can be large. In the example of Figure 11.5, there are 3 colors, 4 items, and 3 sizes resulting in a cube size of $3 \times 4 \times 3 = 36$. Including the summary values, we obtain a

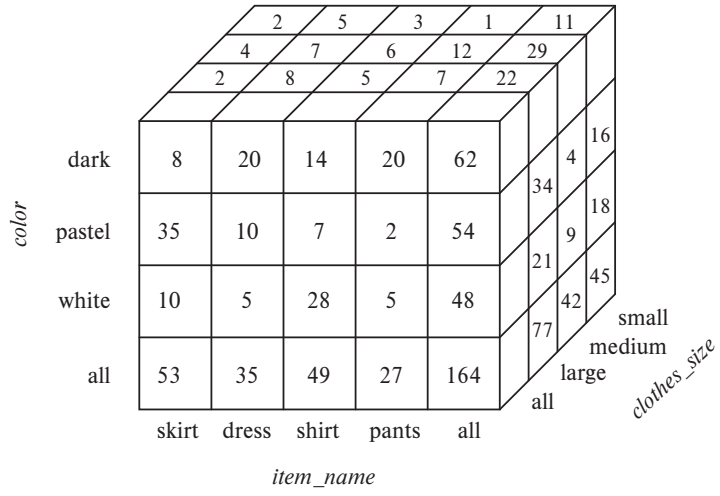


Figure 11.5 Three-dimensional data cube.

$4 \times 5 \times 4$ cube, whose size is 80. In fact, for a table with n dimensions, aggregation can be performed with grouping on each of the 2^n subsets of the n dimensions.[2]

An **online analytic processing** (OLAP) system allows a data analyst to look at different cross-tabs on the same data by *interactively* selecting the attributes in the cross-tab. Each cross-tab is a two-dimensional view on a multidimensional data cube. For instance, the analyst may select a cross-tab on *item_name* and *clothes_size* or a cross-tab on *color* and *clothes_size*. The operation of changing the dimensions used in a cross-tab is called **pivoting**.

OLAP systems allow an analyst to see a cross-tab on *item_name* and *color* for a fixed value of *clothes_size*, for example, large, instead of the sum across all sizes. Such an operation is referred to as **slicing**, since it can be thought of as viewing a slice of the data cube. The operation is sometimes called **dicing**, particularly when values for multiple dimensions are fixed.

When a cross-tab is used to view a multidimensional cube, the values of dimension attributes that are not part of the cross-tab are shown above the cross-tab. The value of such an attribute can be **all**, as shown in Figure 11.4, indicating that data in the cross-tab are a summary over all values for the attribute. Slicing/dicing simply consists of selecting specific values for these attributes, which are then displayed on top of the cross-tab.

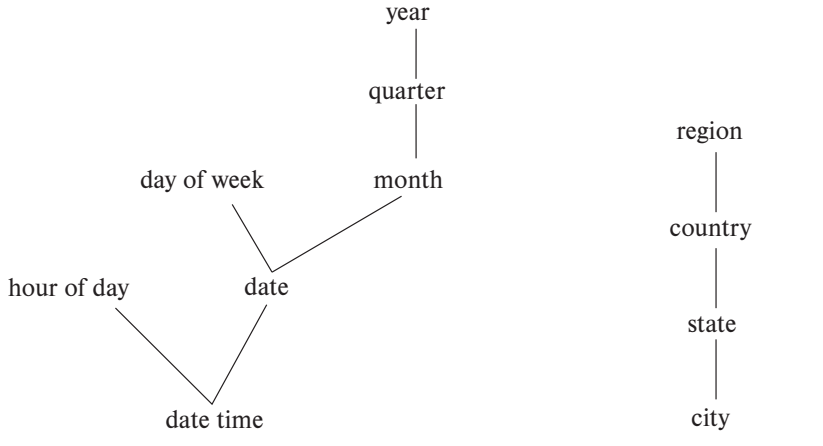
OLAP systems permit users to view data at any desired level of granularity. The operation of moving from finer-granularity data to a coarser granularity (by means of aggregation) is called a **rollup**. In our example, starting from the data cube on the

[2]Grouping on the set of all n dimensions is useful only if the table may have duplicates.

sales table, we got our example cross-tab by rolling up on the attribute *clothes_size*. The opposite operation—that of moving from coarser-granularity data to finer-granularity data—is called a **drill down**. Finer-granularity data cannot be generated from coarse-granularity data; they must be generated either from the original data or from even finer-granularity summary data.

Analysts may wish to view a dimension at different levels of detail. For instance, consider an attribute of type **datetime** that contains a date and a time of day. Using time precise to a second (or less) may not be meaningful: An analyst who is interested in rough time of day may look at only the hour value. An analyst who is interested in sales by day of the week may map the date to a day of the week and look only at that. Another analyst may be interested in aggregates over a month, or a quarter, or for an entire year.

The different levels of detail for an attribute can be organized into a **hierarchy**. Figure 11.6a shows a hierarchy on the **datetime** attribute. As another example, Figure 11.6b shows a hierarchy on location, with the city being at the bottom of the hierarchy, state above it, country at the next level, and region being the top level. In our earlier example, clothes can be grouped by category (for instance, menswear or womenswear); *category* would then lie above *item_name* in our hierarchy on clothes. At the level of actual values, skirts and dresses would fall under the womenswear category and pants and shirts under the menswear category.



(a) time hierarchy

(b) location hierarchy

Figure 11.6 Hierarchies on dimensions.

clothes_size: **all**

category	*item_name*	*color*				
		dark	pastel	white	total	
womenswear	skirt	8	8	10	53	
	dress	20	20	5	35	
	subtotal	28	28	15		88
menswear	pants	14	14	28	49	
	shirt	20	20	5	27	
	subtotal	34	34	33		76
total		62	62	48		164

Figure 11.7 Cross-tabulation of *sales* with hierarchy on *item_name*.

An analyst may be interested in viewing sales of clothes divided as menswear and womenswear, and not interested in individual values. After viewing the aggregates at the level of womenswear and menswear, an analyst may *drill down the hierarchy* to look at individual values. An analyst looking at the detailed level may *drill up the hierarchy* and look at coarser-level aggregates. Both levels can be displayed on the same cross-tab, as in Figure 11.7.

11.3.2 Relational Representation of Cross-Tabs

A cross-tab is different from relational tables usually stored in databases, since the number of columns in the cross-tab depends on the actual data. A change in the data values may result in adding more columns, which is not desirable for data storage. However, a cross-tab view is desirable for display to users. It is straightforward to represent a cross-tab without summary values in a relational form with a fixed number of columns. A cross-tab with summary rows/columns can be represented by introducing a special value **all** to represent subtotals, as in Figure 11.8. The SQL standard actually uses the *null* value in place of **all**, but to avoid confusion with regular null values, we shall continue to use **all**.

Consider the tuples (skirt, **all**, **all**, 53) and (dress, **all**, **all**, 35). We have obtained these tuples by eliminating individual tuples with different values for *color* and *clothes_size*, and by replacing the value of *quantity* with an aggregate—namely, the sum of the quantities. The value **all** can be thought of as representing the set of all values for an attribute. Tuples with the value **all** for the *color* and *clothes_size* dimensions can be obtained by an aggregation on the *sales* relation with a **group by** on the column *item_name*. Similarly, a **group by** on *color*, *clothes_size* can be used to get the tuples with the value **all** for *item_name*, and a **group by** with no attributes (which can simply be omitted in SQL) can be used to get the tuple with value **all** for *item_name*, *color*, and *clothes_size*.

item_name	*color*	*clothes_size*	*quantity*
skirt	dark	**all**	8
skirt	pastel	**all**	35
skirt	white	**all**	10
skirt	**all**	**all**	53
dress	dark	**all**	20
dress	pastel	**all**	10
dress	white	**all**	5
dress	**all**	**all**	35
shirt	dark	**all**	14
shirt	pastel	**all**	7
shirt	white	**all**	28
shirt	**all**	**all**	49
pants	dark	**all**	20
pants	pastel	**all**	2
pants	white	**all**	5
pants	**all**	**all**	27
all	dark	**all**	62
all	pastel	**all**	54
all	white	**all**	48
all	**all**	**all**	164

Figure 11.8 Relational representation of the data in Figure 11.4.

Hierarchies can also be represented by relations. For example, the fact that skirts and dresses fall under the womenswear category and the pants and shirts under the menswear category can be represented by a relation *itemcategory* (*item_name*, *category*). This relation can be joined with the *sales* relation to get a relation that includes the category for each item. Aggregation on this joined relation allows us to get a cross-tab with hierarchy. As another example, a hierarchy on city can be represented by a single relation *city_hierarchy* (ID, *city*, *state*, *country*, *region*), or by multiple relations, each mapping values in one level of the hierarchy to values at the next level. We assume here that cities have unique identifiers, stored in the attribute ID, to avoid confusing between two cities with the same name, for example, the Springfield in Missouri and the Springfield in Illinois.

11.3.3 OLAP in SQL

Analysts using OLAP systems need answers to multiple aggregates to be generated interactively, without having to wait for multiple minutes or hours. This led initially to the development of specialized systems for OLAP (see Note 11.1 on page 535). Many database systems now implement OLAP along with SQL constructs to express OLAP queries. As we saw in Section 5.5.3, several SQL implementations, such as Microsoft

SQL Server and Oracle, support a **pivot** clause that allows creation of cross-tabs. Given the *sales* relation from Figure 11.3, the query:

```
select *
from sales
pivot (
        sum(quantity)
        for color in ('dark','pastel','white')
)
order by item_name;
```

returns the cross-tab shown in Figure 11.9. Note that the **for** clause within the **pivot** clause specifies the *color* values that appear as attribute names in the pivot result. The attribute *color* itself does not appear in the result, although all other attributes are retained, except that the values for the newly created attributes are specified to come from the attribute *quantity*. In case more than one tuple contributes values to a given cell, the aggregate operation within the **pivot** clause specifies how the values should be combined. In the above example, the *quantity* values are summed up.

Note that the pivot clause by itself does not compute the subtotals we saw in the pivot table from Figure 11.4. However, we can first generate the relational representation shown in Figure 11.8, using a cube operation, as outlined shortly, and then apply the pivot clause on that representation to get an equivalent result. In this case, the value **all** must also be listed in the **for** clause, and the **order by** clause needs to be modified to order **all** at the end.

The data in a data cube cannot be generated by a single SQL query if we use only the basic **group by** constructs, since aggregates are computed for several different groupings

item_name	*clothes_size*	*dark*	*pastel*	*white*
dress	small	2	4	2
dress	medium	6	3	3
dress	large	12	3	0
pants	small	14	1	3
pants	medium	6	0	0
pants	large	0	1	2
shirt	small	2	4	17
shirt	medium	6	1	1
shirt	large	6	2	10
skirt	small	2	11	2
skirt	medium	5	9	5
skirt	large	1	15	3

Figure 11.9 Result of SQL pivot operation on the *sales* relation of Figure 11.3.

Note 11.1 OLAP IMPLEMENTATION

The earliest OLAP systems used multidimensional arrays in memory to store data cubes and are referred to as **multidimensional OLAP (MOLAP)** systems. Later, OLAP facilities were integrated into relational systems, with data stored in a relational database. Such systems are referred to as **relational OLAP (ROLAP)** systems. Hybrid systems, which store some summaries in memory and store the base data and other summaries in a relational database, are called **hybrid OLAP (HOLAP)** systems.

Many OLAP systems are implemented as client-server systems. The server contains the relational database as well as any MOLAP data cubes. Client systems obtain views of the data by communicating with the server.

A naïve way of computing the entire data cube (all groupings) on a relation is to use any standard algorithm for computing aggregate operations, one grouping at a time. The naïve algorithm would require a large number of scans of the relation. A simple optimization is to compute an aggregation on, say, (*item_name*, *color*) from an aggregation (*item_name*, *color*, *clothes_size*), instead of from the original relation. The amount of data read drops significantly by computing an aggregate from another aggregate, instead of from the original relation. Further improvements are possible; for instance, multiple groupings can be computed on a single scan of the data.

Early OLAP implementations precomputed and stored entire data cubes, that is, groupings on all subsets of the dimension attributes. Precomputation allows OLAP queries to be answered within a few seconds, even on datasets that may contain millions of tuples adding up to gigabytes of data. However, there are 2^n groupings with n dimension attributes; hierarchies on attributes increase the number further. As a result, the entire data cube is often larger than the original relation that formed the data cube and in many cases it is not feasible to store the entire data cube.

Instead of precomputing and storing all possible groupings, it makes sense to precompute and store some of the groupings, and to compute others on demand. Instead of computing queries from the original relation, which may take a very long time, we can compute them from other precomputed queries. For instance, suppose that a query requires grouping by (*item_name*, *color*), and this has not been precomputed. The query result can be computed from summaries by (*item_name*, *color*, *clothes_size*), if that has been precomputed. See the bibliographical notes for references on how to select a good set of groupings for precomputation, given limits on the storage available for precomputed results.

of the dimension attributes. Using only the basic **group by** construct, we would have to write many separate SQL queries and combine them using a union operation. SQL supports special syntax to allow multiple group by operations to be specified concisely.

As we saw in Section 5.5.4, SQL supports generalizations of the **group by** construct to perform the **cube** and **rollup** operations. The **cube** and **rollup** constructs in the **group by** clause allow multiple **group by** queries to be run in a single query with the result returned as a single relation in a style similar to that of the relation of Figure 11.8.

Consider again our retail shop example and the relation:

$$sales\ (item\_name, color, clothes\_size, quantity)$$

If we want to generate the entire data cube using individual group by queries, we have to write a separate query for each of the following eight sets of group by attributes:

$$\{\ (item\_name, color, clothes\_size), (item\_name, color), (item\_name, clothes\_size),$$
$$(color, clothes\_size), (item\_name), (color), (clothes\_size), ()\ \}$$

where () denotes an empty **group by** list.

As we saw in Section 5.5.4, the **cube** construct allows us to accomplish this in one query:

```
select item_name, color, clothes_size, sum(quantity)
from sales
group by cube(item_name, color, clothes_size);
```

The preceding query produces a relation whose schema is:

$$(item\_name, color, clothes\_size, \textbf{sum}(quantity))$$

So that the result of this query is indeed a relation, tuples in the result contain *null* as the value of those attributes not present in a particular grouping. For example, tuples produced by grouping on *clothes_size* have a schema (*clothes_size*, **sum**(*quantity*)). They are converted to tuples on (*item_name*, *color*, *clothes_size*, **sum**(*quantity*)) by inserting *null* for *item_name* and *color*.

Data cube relations are often very large. The cube query above, with 3 possible colors, 4 possible item names, and 3 sizes, has 80 tuples. The relation of Figure 11.8 is generated by doing a **group by cube** on *item_name* and *color*, with an extra column specified in the **select** clause showing **all** for *clothes_size*.

To generate that relation in SQL, we substitute **all** for *null* using the **grouping** function, as we saw earlier in Section 5.5.4. The **grouping** function distinguishes those nulls generated by OLAP operations from "normal" nulls actually stored in the database or arising from an outer join. Recall that the **grouping** function returns 1 if its argument

is a null value generated by a **cube** or **rollup** and 0 otherwise. We may then operate on the result of a call to **grouping** using **case** expressions to replace OLAP-generated nulls with **all**. Then the relation in Figure 11.8, with occurrences of *null* replaced by **all**, can be computed by the query:

```
select case when grouping(item_name) = 1 then 'all'
            else item_name end as item_name,
       case when grouping(color) = 1 then 'all'
            else color end as color,
       'all' as clothes_size, sum(quantity) as quantity
from sales
group by cube(item_name, color);
```

The **rollup** construct is the same as the **cube** construct except that **rollup** generates fewer **group by** queries. We saw that **group by cube** (*item_name*, *color*, *clothes_size*) generated all eight ways of forming a **group by** query using some (or all or none) of the attributes. In:

```
select item_name, color, clothes_size, sum(quantity)
from sales
group by rollup(item_name, color, clothes_size);
```

the clause **group by rollup**(*item_name*, *color*, *clothes_size*) generates only four groupings:

$$\{ (item\_name, color, clothes\_size), (item\_name, color), (item\_name), () \}$$

Notice that the order of the attributes in the **rollup** makes a difference; the last attribute (*clothes_size*, in our example) appears in only one grouping, the penultimate (second last) attribute in two groupings, and so on, with the first attribute appearing in all groups but one (the empty grouping).

Why might we want the specific groupings that are used in **rollup**? These groups are of frequent practical interest for hierarchies (as in Figure 11.6, for example). For the location hierarchy (*Region, Country, State, City*), we may want to group by *Region* to get sales by region. Then we may want to "drill down" to the level of countries within each region, which means we would group by *Region, Country*. Drilling down further, we may wish to group by *Region, Country, State* and then by *Region, Country, State, City*. The **rollup** construct allows us to specify this sequence of drilling down for further detail.

As we saw earlier in Section 5.5.4, multiple **rollup**s and **cube**s can be used in a single **group by** clause. For instance, the following query:

```
select item_name, color, clothes_size, sum(quantity)
from sales
group by rollup(item_name), rollup(color, clothes_size);
```

generates the groupings:

{ (*item_name*, *color*, *clothes_size*), (*item_name*, *color*), (*item_name*), (*color*, *clothes_size*), (*color*), ( ) }

To understand why, observe that **rollup**(*item_name*) generates two groupings, {(*item_name*), ()}, and **rollup**(*color*, *clothes_size*) generates three groupings, {(*color*, *clothes_size*), (*color*), ( ) }. The Cartesian product of the two gives us the six groupings shown.

Neither the **rollup** nor the **cube** clause gives complete control on the groupings that are generated. For instance, we cannot use them to specify that we want only groupings {(*color*, *clothes_size*), (*clothes_size*, *item_name*)}. Such restricted groupings can be generated by using the **grouping sets** construct, in which one can specify the specific list of groupings to be used. To obtain only groupings {(*color*, *clothes_size*), (*clothes_size*, *item_name*)}, we would write:

select *item_name*, *color*, *clothes_size*, **sum**(*quantity*)
from *sales*
group by grouping sets ((*color*, *clothes_size*), (*clothes_size*, *item_name*));

Specialized languages have been developed for querying multidimensional OLAP schemas, which allow some common tasks to be expressed more easily than with SQL. These include the MDX and DAX query languages developed by Microsoft.

11.3.4 Reporting and Visualization Tools

Report generators are tools to generate human-readable summary reports from a database. They integrate querying the database with the creation of formatted text and summary charts (such as bar or pie charts). For example, a report may show the total sales in each of the past 2 months for each sales region.

The application developer can specify report formats by using the formatting facilities of the report generator. Variables can be used to store parameters such as the month and the year and to define fields in the report. Tables, graphs, bar charts, or other graphics can be defined via queries on the database. The query definitions can make use of the parameter values stored in the variables.

Once we have defined a report structure on a report-generator facility, we can store it and can execute it at any time to generate a report. Report-generator systems provide a variety of facilities for structuring tabular output, such as defining table and column headers, displaying subtotals for each group in a table, automatically splitting long tables into multiple pages, and displaying subtotals at the end of each page.

Figure 11.10 is an example of a formatted report. The data in the report are generated by aggregation on information about orders.

Report-generation tools are available from a variety of vendors, such as SAP Crystal Reports and Microsoft (SQL Server Reporting Services). Several application suites, such as Microsoft Office, provide a way of embedding formatted query results from

Acme Supply Company, Inc.
Quarterly Sales Report

Period: Jan. 1 to March 31, 2009

Region	Category	Sales	Subtotal
North	Computer Hardware	1,000,000	
	Computer Software	500,000	
	All categories		1,500,000
South	Computer Hardware	200,000	
	Computer Software	400,000	
	All categories		600,000
		Total Sales	2,100,000

Figure 11.10 A formatted report.

a database directly into a document. Chart-generation facilities provided by Crystal Reports or by spreadsheets such as Excel can be used to access data from databases and to generate tabular depictions of data or graphical depictions using charts or graphs. Such charts can be embedded within text documents. The charts are created initially from data generated by executing queries against the database; the queries can be re-executed and the charts regenerated when required, to generate a current version of the overall report.

Techniques for **data visualization**, that is, graphical representation of data, that go beyond the basic chart types, are very important for data analysis. Data-visualization systems help users to examine large volumes of data and to detect patterns visually. Visual displays of data—such as maps, charts, and other graphical representations—allow data to be presented compactly to users. A single graphical screen can encode as much information as a far larger number of text screens.

For example, if the user wants to find out whether the occurrence of a disease is correlated to the locations of the patients, the locations of patients can be encoded in a special color—say, red—on a map. The user can then quickly discover locations where problems are occurring. The user may then form hypotheses about why problems are occurring in those locations and may verify the hypotheses quantitatively against the database.

As another example, information about values can be encoded as a color and can be displayed with as little as one pixel of screen area. To detect associations between pairs of items, we can use a two-dimensional pixel matrix, with each row and each column representing an item. The percentage of transactions that buy both items can be encoded by the color intensity of the pixel. Items with high association will show up as bright pixels in the screen—easy to detect against the darker background.

In recent years a number of tools have been developed for web-based data visualization and for the creation of dashboards that display multiple charts showing key organizational information. These include Tableau (www.tableau.com), FusionCharts (www.fusioncharts.com), plotly (plot.ly), Datawrapper (www.datawrapper.de), and Google Charts (developers.google.com/chart), among others. Since their display is based on HTML and JavaScript, they can be used on a wide variety of browsers and on mobile devices.

Interaction is a key element of visualization. For example, a user can "drill down" into areas of interest, such as moving from an aggregate view showing the total sales across an entire year to the monthly sales figures for a particular year. Analysts may wish to interactively add selection conditions to visualize subsets of data. Data visualization tools such as Tableau offer a rich set of features for interactive visualization.

11.4 Data Mining

The term **data mining** refers loosely to the process of analyzing large databases to find useful patterns. Like knowledge discovery in artificial intelligence (also called machine learning) or statistical analysis, data mining attempts to discover rules and patterns from data. However, data mining differs from traditional machine learning and statistics in that it deals with large volumes of data, stored primarily on disk. Today, many machine-learning algorithms also work on very large volumes of data, blurring the distinction between data mining and machine learning. Data-mining techniques form part of the process of **knowledge discovery in databases** (**KDD**).

Some types of knowledge discovered from a database can be represented by a set of **rules**. The following is an example of a rule, stated informally: "Young women with annual incomes greater than $50,000 are the most likely people to buy small sports cars." Of course such rules are not universally true and have degrees of "support" and "confidence," as we shall see. Other types of knowledge are represented by equations relating different variables to each other. More generally, knowledge discovered by applying machine-learning techniques on past instances in a database is represented by a **model**, which is then used for predicting outcomes for new instances. Features or attributes of instances are inputs to the model, and the output of a model is a prediction.

There are a variety of possible types of patterns that may be useful, and different techniques are used to find different types of patterns. We shall study a few examples of patterns and see how they may be automatically derived from a database.

Usually there is a manual component to data mining, consisting of preprocessing data to a form acceptable to the algorithms and post-processing of discovered patterns to find novel ones that could be useful. There may also be more than one type of pattern that can be discovered from a given database, and manual interaction may be needed to pick useful types of patterns. For this reason, data mining is really a semiautomatic process in real life. However, in our description we concentrate on the automatic aspect of mining.

11.4.1 Types of Data-Mining Tasks

The most widely used applications of data mining are those that require some sort of **prediction**. For instance, when a person applies for a credit card, the credit-card company wants to predict if the person is a good credit risk. The prediction is to be based on known attributes of the person, such as age, income, debts, and past debt-repayment history. Rules for making the prediction are derived from the same attributes of past and current credit-card holders, along with their observed behavior, such as whether they defaulted on their credit-card dues. Other types of prediction include predicting which customers may switch over to a competitor (these customers may be offered special discounts to tempt them not to switch), predicting which people are likely to respond to promotional mail ("junk mail"), or predicting what types of phone calling-card usage are likely to be fraudulent.

Another class of applications looks for **associations**, for instance, books that tend to be bought together. If a customer buys a book, an online bookstore may suggest other associated books. If a person buys a camera, the system may suggest accessories that tend to be bought along with cameras. A good salesperson is aware of such patterns and exploits them to make additional sales. The challenge is to automate the process. Other types of associations may lead to discovery of causation. For instance, discovery of unexpected associations between a newly introduced medicine and cardiac problems led to the finding that the medicine may cause cardiac problems in some people. The medicine was then withdrawn from the market.

Associations are an example of **descriptive patterns**. **Clusters** are another example of such patterns. For example, over a century ago a cluster of typhoid cases was found around a well, which led to the discovery that the water in the well was contaminated and was spreading typhoid. Detection of clusters of disease remains important even today.

11.4.2 Classification

Abstractly, the **classification** problem is this: Given that items belong to one of several classes, and given past instances (called **training instances**) of items along with the classes to which they belong, the problem is to predict the class to which a new item belongs. The class of the new instance is not known, so other attributes of the instance must be used to predict the class.

As an example, suppose that a credit-card company wants to decide whether or not to give a credit card to an applicant. The company has a variety of information about the person, such as her age, educational background, annual income, and current debts, that it can use for making a decision.

To make the decision, the company assigns a credit worthiness level of excellent, good, average, or bad to each of a sample set of current or past customers according to each customer's payment history. These instances form the set of training instances.

Then, the company attempts to learn rules or models that classify the credit-worthiness of a new applicant as excellent, good, average, or bad, on the basis of the

information about the person, other than the actual payment history (which is unavailable for new customers). There are a number of techniques for classification, and we outline a few of them in this section.

11.4.2.1 Decision-Tree Classifiers

Decision-tree classifiers are a widely used technique for classification. As the name suggests, **decision-tree classifiers** use a tree; each leaf node has an associated class, and each internal node has a predicate (or more generally, a function) associated with it. Figure 11.11 shows an example of a decision tree. To keep the example simple, we use just two attributes: education level (highest degree earned) and income.

To classify a new instance, we start at the root and traverse the tree to reach a leaf; at an internal node we evaluate the predicate (or function) on the data instance to find which child to go to. The process continues until we reach a leaf node. For example, if the degree level of a person is masters, and the person's income is 40K, starting from the root we follow the edge labeled "masters," and from there the edge labeled "25K to 75K," to reach a leaf. The class at the leaf is "good," so we predict that the credit risk of that person is good.

There are a number of techniques for building decision-tree classifiers from a given training set. We omit details, but you can learn more details from the references provided in the Further Reading section.

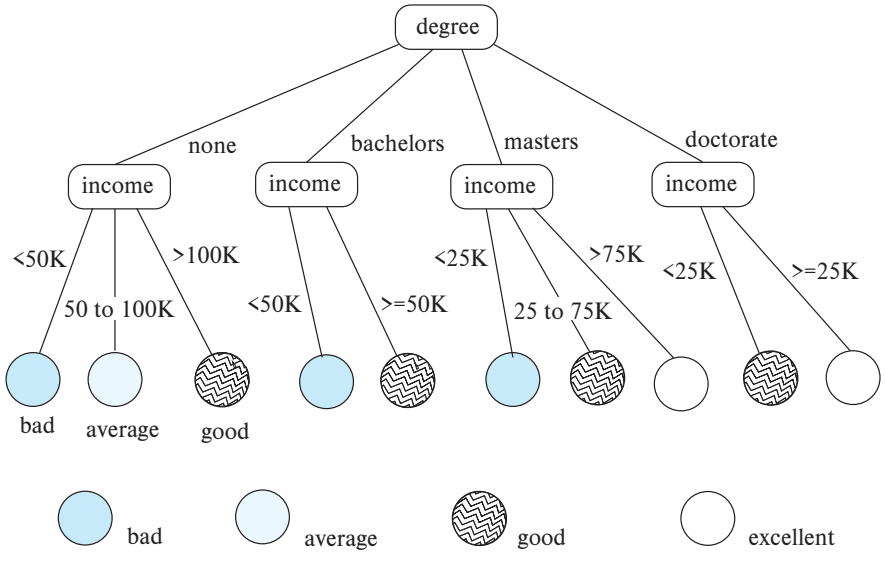


Figure 11.11 Classification tree.

11.4.2.2 Bayesian Classifiers

Bayesian classifiers find the distribution of attribute values for each class in the training data; when given a new instance d, they use the distribution information to estimate, for each class c_j, the probability that instance d belongs to class c_j, denoted by $p(c_j|d)$, in a manner outlined here. The class with maximum probability becomes the predicted class for instance d.

To find the probability $p(c_j|d)$ of instance d being in class c_j, Bayesian classifiers use **Bayes' theorem**, which says:

$$p(c_j|d) = \frac{p(d|c_j)p(c_j)}{p(d)}$$

where $p(d|c_j)$ is the probability of generating instance d given class c_j, $p(c_j)$ is the probability of occurrence of class c_j, and $p(d)$ is the probability of instance d occurring. Of these, $p(d)$ can be ignored since it is the same for all classes. $p(c_j)$ is simply the fraction of training instances that belong to class c_j.

For example, let us consider a special case where only one attribute, *income*, is used for classification, and suppose we need to classify a person whose income is 76,000. We assume that income values are broken up into buckets, and we assume that the bucket containing 76,000 contains values in the range (75,000, 80,000). Suppose among instances of class *excellent*, the probability of income being in (75,000, 80,000) is 0.1, while among instances of class *good*, the probability of income being in (75,000, 80,000) is 0.05. Suppose also that overall 0.1 fraction of people are classified as *excellent*, and 0.3 are classified as *good*. Then, $p(d|c_j)p(c_j)$ for class *excellent* is .01, while for class *good*, it is 0.015. The person would therefore be classified in class *good*.

In general, multiple attributes need to be considered for classification. Then, finding $p(d|c_j)$ exactly is difficult, since it requires the distribution of instances of c_j, across all combinations of values for the attributes used for classification. The number of such combinations (for example of income buckets, with degree values and other attributes) can be very large. With a limited training set used to find the distribution, most combinations would not have even a single training set matching them, leading to incorrect classification decisions. To avoid this problem, as well as to simplify the task of classification, **naive Bayesian classifiers** assume attributes have independent distributions and thereby estimate:

$$p(d|c_j) = p(d_1|c_j) * p(d_2|c_j) * \cdots * p(d_n|c_j)$$

That is, the probability of the instance d occurring is the product of the probability of occurrence of each of the attribute values d_i of d, given the class is c_j.

The probabilities $p(d_i|c_j)$ derive from the distribution of values for each attribute i, for each class c_j. This distribution is computed from the training instances that belong to each class c_j; the distribution is usually approximated by a histogram. For instance, we may divide the range of values of attribute i into equal intervals, and store the fraction of instances of class c_j that fall in each interval. Given a value d_i for attribute i, the

value of $p(d_i|c_j)$ is simply the fraction of instances belonging to class c_j that fall in the interval to which d_i belongs.

11.4.2.3 Support Vector Machine Classifiers

The **Support Vector Machine** (SVM) is a type of classifier that has been found to give very accurate classification across a range of applications. We provide some basic information about Support Vector Machine classifiers here; see the references in the bibliographical notes for further information.

Support Vector Machine classifiers can best be understood geometrically. In the simplest case, consider a set of points in a two-dimensional plane, some belonging to class *A*, and some belonging to class *B*. We are given a training set of points whose class (*A* or *B*) is known, and we need to build a classifier of points using these training points. This situation is illustrated in Figure 11.12, where the points in class *A* are denoted by X marks, while those in class *B* are denoted by O marks.

Suppose we can draw a line on the plane, such that all points in class *A* lie to one side and all points in line *B* lie to the other. Then, the line can be used to classify new points, whose class we don't already know. But there may be many possible such lines that can separate points in class *A* from points in class *B*. A few such lines are shown in Figure 11.12. The Support Vector Machine classifier chooses the line whose distance from the nearest point in either class (from the points in the training dataset) is maximum. This line (called the *maximum margin line*) is then used to classify other points into class *A* or *B*, depending on which side of the line they lie on. In Figure 11.12, the maximum margin line is shown in bold, while the other lines are shown as dashed lines.

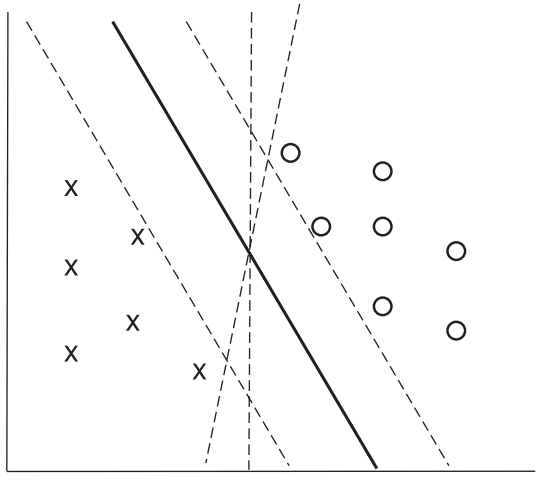


Figure 11.12 Example of a Support Vector Machine classifier.

The preceding intuition can be generalized to more than two dimensions, allowing multiple attributes to be used for classification; in this case, the classifier finds a dividing plane, not a line. Further, by first transforming the input points using certain functions, called *kernel functions*, Support Vector Machine classifiers can find nonlinear curves separating the sets of points. This is important for cases where the points are not separable by a line or plane. In the presence of noise, some points of one class may lie in the midst of points of the other class. In such cases, there may not be any line or meaningful curve that separates the points in the two classes; then, the line or curve that most accurately divides the points into the two classes is chosen.

Although the basic formulation of Support Vector Machines is for binary classifiers, i.e., those with only two classes, they can be used for classification into multiple classes as follows: If there are N classes, we build N classifiers, with classifier i performing a binary classification, classifying a point either as in class i or not in class i. Given a point, each classifier i also outputs a value indicating how related a given point is to class i. We then apply all N classifiers on a given point and choose the class for which the relatedness value is the highest.

11.4.2.4 Neural Network Classifiers

Neural-net classifiers use the training data to train artificial neural nets. There is a large body of literature on neural nets; we do not provide details here, but we outline a few key properties of neural network classifiers.

Neural networks consist of several layers of "neurons," each of which are connected to neurons in the preceding layer. An input instance of the problem is fed to the first layer; neurons at each layer are "activated" based on some function applied to the inputs at the preceding layer. The function applied at each neuron computes a weighted combination of the activations of the input neurons and generates an output based on the weighted combination. The activation of a neuron in one layer thus affects the activation of neurons in the next layer. The final output layer typically has one neuron corresponding to each class of the classification problem being addressed. The neuron with maximum activation for a given input decides the predicted class for that input.

Key to the success of a neural network is the weights used in the computation described above. These weights are learned, based on training data. They are initially set to some default value, and then training data are used to learn the weights. Training is typically done by applying each input to the current state of the neural network and checking if the prediction is correct. If not, a *backpropagation algorithm* is used to tweak the weights of the neurons in the network, to bring the prediction closer to the correct one for the current input. Repeating this process results in a trained neural network, which can then be used for classification on new inputs.

In recent years, neural networks have achieved a great degree of success for tasks which were earlier considered very hard, such as vision (e.g., recognition of objects in images), speech recognition, and natural language translation. A simple example of a vision task is that of identifying the species, such as cat or dog, given an image of

an animal; such problems are basically classification problems. Other examples include identifying object occurrences in an image and assigning a class label to each identified object.

Deep neural networks, which are neural networks with a large number of layers, have proven very successful at such tasks, if given a very large number of training instances. The term **deep learning** refers to the machine-learning techniques that create such deep neural networks, and train them on very large numbers of training instances.

11.4.3 Regression

Regression deals with the prediction of a value, rather than a class. Given values for a set of variables, $X_1, X_2, \ldots, X_n$, we wish to predict the value of a variable Y. For instance, we could treat the level of education as a number and income as another number, and, on the basis of these two variables, we wish to predict the likelihood of default, which could be a percentage chance of defaulting, or the amount involved in the default.

One way is to infer coefficients $a_0, a_1, a_2, \ldots, a_n$ such that:

$$Y = a_0 + a_1 * X_1 + a_2 * X_2 + \cdots + a_n * X_n$$

Finding such a linear polynomial is called **linear regression**. In general, we wish to find a curve (defined by a polynomial or other formula) that fits the data; the process is also called **curve fitting**.

The fit may be only approximate, because of noise in the data or because the relationship is not exactly a polynomial, so regression aims to find coefficients that give the best possible fit. There are standard techniques in statistics for finding regression coefficients. We do not discuss these techniques here, but the bibliographical notes provide references.

11.4.4 Association Rules

Retail shops are often interested in associations between different items that people buy. Examples of such associations are:

- Someone who buys bread is quite likely also to buy milk.
- A person who bought the book *Database System Concepts* is quite likely also to buy the book *Operating System Concepts*.

Association information can be used in several ways. When a customer buys a particular book, an online shop may suggest associated books. A grocery shop may decide to place bread close to milk, since they are often bought together, to help shoppers finish their task faster. Or, the shop may place them at opposite ends of a row and place other associated items in between to tempt people to buy those items as well as the shoppers walk from one end of the row to the other. A shop that offers discounts on one associ-

ated item may not offer a discount on the other, since the customer will probably buy the other anyway.

An example of an association rule is:

$$bread \Rightarrow milk$$

In the context of grocery-store purchases, the rule says that customers who buy bread also tend to buy milk with a high probability. An association rule must have an associated **population**: The population consists of a set of **instances**. In the grocery-store example, the population may consist of all grocery-store purchases; each purchase is an instance. In the case of a bookstore, the population may consist of all people who made purchases, regardless of when they made a purchase. Each customer is an instance. In the bookstore example, the analyst has decided that when a purchase is made is not significant, whereas for the grocery-store example, the analyst may have decided to concentrate on single purchases, ignoring multiple visits by the same customer.

Rules have an associated *support*, as well as an associated *confidence*. These are defined in the context of the population:

- **Support** is a measure of what fraction of the population satisfies both the antecedent and the consequent of the rule.

  For instance, suppose only 0.001 percent of all purchases include milk and screwdrivers. The support for the rule:

  $$milk \Rightarrow screwdrivers$$

  is low. The rule may not even be statistically significant—perhaps there was only a single purchase that included both milk and screwdrivers. Businesses are usually not interested in rules that have low support, since they involve few customers and are not worth bothering about.

  On the other hand, if 50 percent of all purchases involve milk and bread, then support for rules involving bread and milk (and no other item) is relatively high, and such rules may be worth attention. Exactly what minimum degree of support is considered desirable depends on the application.

- **Confidence** is a measure of how often the consequent is true when the antecedent is true. For instance, the rule:

  $$bread \Rightarrow milk$$

  has a confidence of 80 percent if 80 percent of the purchases that include bread also include milk. A rule with a low confidence is not meaningful. In business applications, rules usually have confidences significantly less than 100 percent, whereas in other domains, such as in physics, rules may have high confidences.

  Note that the confidence of $bread \Rightarrow milk$ may be very different from the confidence of $milk \Rightarrow bread$, although both have the same support.

11.4.5 Clustering

Intuitively, clustering refers to the problem of finding clusters of points in the given data. The problem of **clustering** can be formalized from distance metrics in several ways. One way is to phrase it as the problem of grouping points into k sets (for a given k) so that the average distance of points from the *centroid* of their assigned cluster is minimized. [3] Another way is to group points so that the average distance between every pair of points in each cluster is minimized. There are other definitions too; see the bibliographical notes for details. But the intuition behind all these definitions is to group similar points together in a single set.

Another type of clustering appears in classification systems in biology. (Such classification systems do not attempt to *predict* classes; rather they attempt to cluster related items together.) For instance, leopards and humans are clustered under the class mammalia, while crocodiles and snakes are clustered under reptilia. Both mammalia and reptilia come under the common class chordata. The clustering of mammalia has further subclusters, such as carnivora and primates. We thus have **hierarchical clustering**. Given characteristics of different species, biologists have created a complex hierarchical clustering scheme grouping related species together at different levels of the hierarchy.

The statistics community has studied clustering extensively. Database research has provided scalable clustering algorithms that can cluster very large datasets (that may not fit in memory).

An interesting application of clustering is to predict what new movies (or books or music) a person is likely to be interested in on the basis of:

1. The person's past preferences in movies.
2. Other people with similar past preferences.
3. The preferences of such people for new movies.

One approach to this problem is as follows: To find people with similar past preferences we create clusters of people based on their preferences for movies. The accuracy of clustering can be improved by previously clustering movies by their similarity, so even if people have not seen the same movies, if they have seen similar movies they would be clustered together. We can repeat the clustering, alternately clustering people, then movies, then people, and so on until we reach an equilibrium. Given a new user, we find a cluster of users most similar to that user, on the basis of the user's preferences for movies already seen. We then predict movies in movie clusters that are popular with that user's cluster as likely to be interesting to the new user. In fact, this problem

[3] The centroid of a set of points is defined as a point whose coordinate on each dimension is the average of the coordinates of all the points of that set on that dimension. For example, in two dimensions, the centroid of a set of points $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ is given by $\left(\frac{\sum_{i=1}^{n} x_i}{n}, \frac{\sum_{i=1}^{n} y_i}{n}\right)$.

is an instance of *collaborative filtering*, where users collaborate in the task of filtering information to find information of interest.

11.4.6 Text Mining

Text mining applies data-mining techniques to textual documents. There are a number of different text mining tasks. One such task is **sentiment analysis**. For example, suppose a company wishes to find out how users have reacted to a new product. There are typically a large number of product reviews on the web—for example, reviews by different users on e-commerce platforms. Reading each review to find out reactions is not practical for a human. Instead, the company may analyze reviews to find the *sentiment* of the reviews of the product; the sentiment could be positive, negative, or neutral. The occurrence of specific words such as excellent, good, awesome, beautiful, and so on are correlated with a positive sentiment, while words such as awful, average, worthless, poor quality, and so on are correlated with a negative sentiment. Sentiment analysis techniques can be used to analyze the reviews and come up with an overall score reflecting the broad sense of the reviews.

Another task is **information extraction**, which creates structured information from unstructured textual descriptions, or semi-structured data such as tabular displays of data in documents. A key subtask of this process is **entity recognition**, that is, the task of identifying mentions of entities in text and disambiguating them. For example, an article may mention the name Michael Jordan. There are at least two famous people named Michael Jordan: one was a basketball player, while the other is a professor who is a well known machine-learning expert. **Disambiguation** is the process of figuring out which of these two is being referred to in a particular article, and it can be done based on the article context; in this case, an occurrence of the name Michael Jordan in a sports article probably refers to the basketball player, while an occurrence in a machine-learning paper probably refers to the professor. After entity recognition, other techniques may be used to learn attributes of entities and to learn relationships between entities.

Information extraction can be used in many ways. For example, it can be used to analyze customer support conversations or reviews posted on social media, to judge customer satisfaction, and to decide when intervention is needed to retain customers. Service providers may want to know what aspect of the service such as pricing, quality, hygiene, or behavior of the person providing the service, a review was positive or negative about; information extraction techniques can be used to infer what aspect an article or a part of an article is about, and to infer the associated sentiment. Attributes such as the location of service can also be extracted and are important for taking corrective action.

Information extracted from the enormous collection of documents and other resources on the web can be valuable for many tasks. Such extracted information can be represented in a graph, called a **knowledge graph**, which we outlined in Section 8.1.4. Such knowledge graphs are used by web search engines to generate more meaningful answers to user queries.

11.5 Summary

- Data analytics systems analyze online data collected by transaction-processing systems, along with data collected from other sources, to help people make business decisions. Decision-support systems come in various forms, including OLAP systems and data-mining systems.
- Data warehouses help gather and archive important operational data. Warehouses are used for decision support and analysis on historical data, for instance, to predict trends. Data cleansing from input data sources is often a major task in data warehousing. Warehouse schemas tend to be multidimensional, involving one or a few very large fact tables and several much smaller dimension tables.
- Online analytical processing (OLAP) tools help analysts view data summarized in different ways, so that they can gain insight into the functioning of an organization.
    - OLAP tools work on multidimensional data, characterized by dimension attributes and measure attributes.
    - The data cube consists of multidimensional data summarized in different ways. Precomputing the data cube helps speed up queries on summaries of data.
    - Cross-tab displays permit users to view two dimensions of multidimensional data at a time, along with summaries of the data.
    - Drill down, rollup, slicing, and dicing are among the operations that users perform with OLAP tools.
- The SQL standard provides a variety of operators for data analysis, including **cube**, **rollup**, and **pivot** operations.
- Data mining is the process of semiautomatically analyzing large databases to find useful patterns. There are a number of applications of data mining, such as prediction of values based on past examples, finding of associations between purchases, and automatic clustering of people and movies.
- Classification deals with predicting the class of test instances by using attributes of the test instances, based on attributes of training instances, and the actual class of training instances. There are several types of classifiers, such as:
    - Decision-tree classifiers, which perform classification by constructing a tree based on training instances with leaves having class labels.
    - Bayesian classifiers, which are based on probability theory.
    - The support vector machine is another widely used classification technique.
    - Neural networks, and in particular deep learning, has been very successful in classification and related tasks in the context of vision, speech recognition, and language understanding and translation.

- Association rules identify items that co-occur frequently, for instance, items that tend to be bought by the same customer. Correlations look for deviations from expected levels of association.
- Other types of data mining include clustering and text mining.

Review Terms

- Decision-support systems
- Business intelligence
- Data warehousing
  - Gathering data
  - Source-driven architecture
  - Destination-driven architecture
  - Data cleansing
  - Extract, transform, load (ETL)
  - Extract, load, transform (ELT)
- Warehouse schemas
  - Fact table
  - Dimension tables
  - Star schema
  - Snowflake schema
- Column-oriented storage
- Online analytical processing (OLAP)
- Multidimensional data
  - Measure attributes
  - Dimension attributes
  - Hierarchy
  - Cross-tabulation / Pivoting
  - Data cube
  - Rollup and drill down
  - SQL **group by cube**, **group by rollup**
- Data visualization
- Data mining
- Prediction
- Classification
  - Training data
  - Test data
- Decision-tree classifiers
- Bayesian classifiers
  - Bayes' theorem
  - Naive Bayesian classifiers
- Support Vector Machine (SVM)
- Regression
- Neural-networks
- Deep learning
- Association rules
- Clustering
- Text mining
  - Sentiment analysis
  - Information extraction
  - Named entity recognition
  - Knowledge graph

Practice Exercises

11.1 Describe benefits and drawbacks of a source-driven architecture for gathering of data at a data warehouse, as compared to a destination-driven architecture.

11.2 Draw a diagram that shows how the *classroom* relation of our university example as shown in Appendix A would be stored under a column-oriented storage structure.

11.3 Consider the *takes* relation. Write an SQL query that computes a cross-tab that has a column for each of the years 2017 and 2018, and a column for **all**, and one row for each course, as well as a row for **all**. Each cell in the table should contain the number of students who took the corresponding course in the corresponding year, with column **all** containing the aggregate across all years, and row **all** containing the aggregate across all courses.

11.4 Consider the data warehouse schema depicted in Figure 11.2. Give an SQL query to summarize sales numbers and price by store and date, along with the hierarchies on store and date.

11.5 Classification can be done using *classification rules*, which have a *condition*, a *class*, and a *confidence*; the confidence is the percentage of the inputs satisfying the condition that fall in the specified class.

For example, a classification rule for credit ratings may have a condition that salary is between $30,000 and $50,000, and education level is graduate, with the credit rating class of *good*, and a confidence of 80%. A second rule may have a condition that salary is between $30,000 and $50,000, and education level is high-school, with the credit rating class of *satisfactory*, and a confidence of 80%. A third rule may have a condition that salary is above $50,001, with the credit rating class of *excellent*, and confidence of 90%. Show a decision tree classifier corresponding to the above rules.

Show how the decision tree classifier can be extended to record the confidence values.

11.6 Consider a classification problem where the classifier predicts whether a person has a particular disease. Suppose that 95% of the people tested do not suffer from the disease. Let *pos* denote the fraction of *true positives*, which is 5% of the test cases, and let *neg* denote the fraction of *true negatives*, which is 95% of the test cases. Consider the following classifiers:

- Classifier C_1, which always predicts negative (a rather useless classifier, of course).
- Classifier C_2, which predicts positive in 80% of the cases where the person actually has the disease but also predicts positive in 5% of the cases where the person does not have the disease.

- Classifier C_3, which predicts positive in 95% of the cases where the person actually has the disease but also predicts positive in 20% of the cases where the person does not have the disease.

For each classifier, let *t_pos* denote the *true positive* fraction, that is the fraction of cases where the classifier prediction was positive, and the person actually had the disease. Let *f_pos* denote the *false positive* fraction, that is the fraction of cases where the prediction was positive, but the person did not have the disease. Let *t_neg* denote *true negative* and *f_neg* denote *false negative* fractions, which are defined similarly, but for the cases where the classifier prediction was negative.

a. Compute the following metrics for each classifier:
   i. *Accuracy*, defined as $(t\_pos + t\_neg)/(pos+neg)$, that is, the fraction of the time when the classifier gives the correct classification.
   ii. *Recall* (also known as *sensitivity*) defined as t_pos/pos, that is, how many of the actual positive cases are classified as positive.
   iii. *Precision*, defined as $t\_pos/(t\_pos+f\_pos)$, that is, how often the positive prediction is correct.
   iv. *Specificity*, defined as t_neg/neg.

b. If you intend to use the results of classification to perform further screening for the disease, how would you choose between the classifiers?

c. On the other hand, if you intend to use the result of classification to start medication, where the medication could have harmful effects if given to someone who does not have the disease, how would you choose between the classifiers?

Exercises

11.7 Why is column-oriented storage potentially advantageous in a database system that supports a data warehouse?

11.8 Consider each of the *takes* and *teaches* relations as a fact table; they do not have an explicit measure attribute, but assume each table has a measure attribute *reg_count* whose value is always 1. What would the dimension attributes and dimension tables be in each case. Would the resultant schemas be star schemas or snowflake schemas?

11.9 Consider the star schema from Figure 11.2. Suppose an analyst finds that monthly total sales (sum of the *price* values of all *sales* tuples) have decreased, instead of growing, from April 2018 to May 2018. The analyst wishes to check if there are specific item categories, stores, or customer countries that are responsible for the decrease.

a. What are the aggregates that the analyst would start with, and what are the relevant drill-down operations that the analyst would need to execute?

b. Write an SQL query that shows the item categories that were responsible for the decrease in sales, ordered by the impact of the category on the sales decrease, with categories that had the highest impact sorted first.

11.10 Suppose half of all the transactions in a clothes shop purchase jeans, and one-third of all transactions in the shop purchase T-shirts. Suppose also that half of the transactions that purchase jeans also purchase T-shirts. Write down all the (nontrivial) association rules you can deduce from the above information, giving support and confidence of each rule.

11.11 The organization of parts, chapters, sections, and subsections in a book is related to clustering. Explain why, and to what form of clustering.

11.12 Suggest how predictive mining techniques can be used by a sports team, using your favorite sport as an example.

Tools

Data warehouse systems are available from Teradata, Teradata Aster, SAP IQ (formerly known as Sybase IQ), and Amazon Redshift, all of which support parallel processing across a large number of machines. A number of databases including Oracle, SAP HANA, Microsoft SQL Server, and IBM DB2 support data warehouse applications by adding features such as columnar storage. There are a number of commercial ETL tools including tools from Informatica, Business Objects, IBM InfoSphere, Microsoft Azure Data Factory, Microsoft SQL Server Integration Services, Oracle Warehouse Builder, and Pentaho Data Integration. Open-source ETL tools include Apache NiFi (nifi.apache.org), Jasper ETL (www.jaspersoft.com/data-integration) and Talend (sourceforge.net/projects/talend-studio). Apache Kafka (kafka.apache.org) can also be used to build ETL systems.

Most database vendors provide OLAP tools as part of their database systems, or as add-on applications. These include OLAP tools from Microsoft Corp., Oracle, IBM and SAP. The Mondrian OLAP server (github.com/pentaho/mondrian) is an open-source OLAP server. Apache Kylin (kylin.apache.org) is an open-source distributed analytics engine which can process data stored in Hadoop, build OLAP cubes and store them in the HBase key-value store, and then query the stored cubes using SQL. Many companies also provide analysis tools for specific applications, such as customer relationship management.

Tools for visualization include Tableau (www.tableau.com), FusionCharts (www.fusioncharts.com), plotly (plot.ly), Datawrapper (www.datawrapper.de), and Google Charts (developers.google.com/chart).

The Python language is very popular for machine-learning tasks, due to the availability of a number of open-source libraries for machine-learning tasks. The R language is also used widely for statistical analysis and machine learning, for the same reasons. Popular utility libraries in Python include are NumPy (www.numpy.org) which provides operations on arrays and matrices, SciPy (www.scipy.org), which provides linear algebra, optimization and statistics functions, and Pandas (pandas.pydata.org), which provides a relational abstraction of data. Popular machine-learning libraries in Python include SciKit-Learn (scikit-learn.org), which adds image-processing and machine-learning functionality to SciPy. Deep learning libraries in Python include Keras (keras.io), and TensorFlow (www.tensorflow.org) which was developed by Google; TensorFlow provides APIs in several languages, with particularly good support for Python. Text mining is supported by natural language processing libraries, such as NLTK (www.nltk.org) and web crawling libraries, such as Scrapy (scrapy.org). Visualization is supported by libraries such as Matplotlib (matplotlib.org), Plotly (plot.ly) and Bokeh (bokeh.pydata.org).

Open-source tools for data mining include RapidMiner (rapidminer.com), Weka (www.cs.waikato.ac.nz/ml/weka), and Orange (orange.biolab.si). Commercial tools include SAS Enterprise Miner, IBM Intelligent Miner, and Oracle Data Mining.

Further Reading

[Kimball et al. (2008)] and [Kimball and Ross (2013)] provide textbook coverage of data warehouses and multidimensional modeling.

[Mitchell (1997)] is a classic textbook on machine learning and covers classification techniques in detail. [Goodfellow et al. (2016)] is a definitive text on deep learning. [Witten et al. (2011)] and [Han et al. (2011)] provide textbook coverage of data mining. [Agrawal et al. (1993)] introduced the notion of association rules.

Information about the R language and environment may be found at www.r-project.org; information about the SparkR package, which provides an R frontend to Apache Spark, may be found at spark.apache.org/docs/latest/sparkr.html.

[Chakrabarti (2002)], [Manning et al. (2008)] and [Baeza-Yates and Ribeiro-Neto (2011)] provide textbook description of information retrieval, including extensive coverage of data-mining tasks related to textual and hypertext data, such as classification and clustering.

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